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ALUMINUM AND MAGNESIUM PRODUCTS PREPARATION FOR SHIPMENT AND STORAGE

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DEPARTMENT OF DEFENSE
WASHINGTON, D.C.

ALUMINUM AND MAGNESIUM PRODUCTS
PREPARATION FOR SHIPMENT AND STORAGE

MIL-STD-649A

1. This standard has been approved by the Department of Defense and is mandatory for use by the Army, The Navy, and the Air Force, effective 12 July 1967.

2. Recommended corrections, additions, or deletions should be addressed to: Headquarters, DSA, Standardization Division, Washington, D. C. 20230.

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CONTENTS

		Page
1.	INTRODUCTION AND SCOPE	1
1.1	Scope	1
2.	REFERENCED DOCUMENTS	1
2.1	Other Publications	2
3.	DEFINITIONS	2
3.1	Preservation and Packaging	2
3.2	Packing	2
3.3	Levels of Preservation and Packaging	2
3.3.1	Level A	2
3.3.2	Level C	2
3.4	Levels of Packing	2
3.4.1	Level A	2
3.4.2	Level B	2
3.4.3	Level C	2
3.5	Definition of Terms	3
4.	GENERAL REQUIREMENTS	3
4.1	Materials, Methods and Containers	3
4.1.1	Non Specification Materials	4
4.1.2	Non Specification Packaging and Packing Procedures	4
4.1.3	Inside Container Dimensions	4
5.	DETAIL REQUIREMENTS	4
5.1	Preservation	4
5.1.1	Cleaning	4
5.1.2	Preservatives	4
5.1.3	Surfaces Shall be Preserved Unless Otherwise Specified	4
5.1.4	Magnesium Sheet and Plate	4
5.1.5	Application of Preservative Coating	4
5.2	Preservation and Packaging	4
5.2.1	Level A	4
5.2.2	Level C	4
5.3	Packing	4
5.3.1	Level A	4
5.3.2	Level B	4
5.3.3	Level C	4
5.4	Materials	4
5.4.1	Fiberboard	4
5.4.2	Paper and Barrier Materials	5
5.4.3	Wood	5
5.4.4	Lumber	5
5.4.5	Plywood and Paper Overlaid Veneers	5
5.4.5.1	Non Specification Plywood	5
5.4.5.2	Non Specification Paper Overlaid Veneer	5
5.4.6	Nails and Staples	5
5.4.7	Strapping	5
5.4.8	Tapes	5
5.5	Containers and Methods	5
5.5.1	Boxes	5
5.5.1.1	Boxes for Flat Sheet and Plate	5
5.5.1.2	Sheet or Plate Boxes on Skids	5
5.5.1.3	Two or More Containers Weighing Less than 600 Pounds	5
5.5.1.4	Two to Ten Boxes Not Exceeding 6000 Pounds Gross	5
5.5.1.5	Double Wall or Triple Wall Fiberboard or Wirebound Boxes	6
5.5.1.6	Boxes for Extruded and Structural Shapes	6

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5.5.1.7	Boxes for Products Other Than Extruded or Structural Shapes, Flat Sheet and Plate	6
5.5.1.8	Requirements for Wood, Plywood and Paper Overlaid Veneer Boxes	6
5.5.1.8.1	Ends	6
5.5.1.8.2	End Cleats	6
5.5.1.8.3	Splices	6
5.5.1.8.4	Additional Cleats	6
5.5.1.8.5	Spacing of Additional Cleats	6
5.5.1.9	Requirements for Wirebound Boxes of Flat Sheet and Plate	6
5.5.1.10	Requirements for Double Wall and Triple Wall Fiberboard Boxes for Flat Sheet and Plate	6
5.5.1.10.1	Double Wall or Triple Wall Fiberboard Boxes Weighing over 300 Pounds	6
5.5.1.11	Palletized Shipment	6
5.5.2	Requirements for All Boxes	6
5.5.2.1	Blocking and Bracing	7
5.5.2.2	Box Skids	7
5.5.2.3	Nailing and Stapling	7
5.5.2.4	Strapping	7
5.5.2.4.1	Strap Spacing	7
5.5.3	Skid Systems	7
5.5.3.1	Skid Ends	7
5.5.3.2	Lengthwise Skids	7
5.5.3.3	Crosswise Skids	7
5.5.3.4	Lengthwise Skids with Crossmembers	7
5.5.3.5	Deckboards	7
5.5.3.5.1	Nailing for Deckboards	7
5.5.3.6	Sides, Ends and Tops	7
5.5.3.7	End Cleats	8
5.5.3.8	Top Battens	8
5.5.3.9	Nailing for Skid Systems	8
5.5.3.10	Strapping for Skids	8
5.5.4	Bundles	8
5.5.4.1	Wrapping	8
5.5.4.2	Strapping for Bundles and Coils	8
5.5.5	Crates Construction Details	8
5.5.5.1	Style 1	8
5.5.5.2	Style 2	8
5.5.5.3	Style 1 Crates	8
5.5.5.3.1	Fabrication	8
5.5.5.3.2	Nails	8
5.5.5.3.3	Strapping	8
5.5.5.3.3.1	Strap Spacing	8
5.5.5.4	Style 2 Crates	8
5.5.5.4.1	Fabrication	8
5.5.5.4.2	Skids	8
5.5.5.4.3	Deckboards	9
5.5.5.4.4	Battens	9
5.5.5.4.5	Framing Members	9
5.5.5.4.6	Sling Reinforcements	9
5.5.5.4.7	Ends	9
5.5.5.4.8	Nails	9
5.5.5.4.9	Strapping	9
5.5.5.4.9.1	Strap Spacing	9
5.5.6	Reels	9
5.5.7	Spools	9
5.5.8	Steel Pails	9
5.5.9	Steel Drums	9
5.5.10	Metal Cans	9
5.5.11	Fiber Drums	9
5.5.12	Special Containers	9
5.5.12.1	Net Weight per Container	9
5.5.12.2	When Weight Exceeds Those Specified in Table XIII	9

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5.5.12.3	When Weight is Less Than Minimum Specified in Table XIII	10
5.5.12.4	Quantity of Flat Sheet or Plate in Individual Container Shall Conform to Unit Pack Requirements of Table IIIb	10
5.6	Markings for Shipment	10
5.6.1	When Order Requests Two Units Shipments and Markings Shall Include Quantity of Both Units	10

LIST OF FIGURES

FIGURE		Page
1	Style 1 Box	11
2	Style 2 Box	11
3	Style 4 Box	11
4	Style 5 Box	11
5	Style 1 Box for Bar, Rod, Shapes, Tubular Products and Wire	12
6	Style 4 Box for Bar, Rod, Shapes, Tubular Products and Wire	12
7	Style 2 Box for Bar, Rod, Shapes, Tubular Products and Wire	12
8	Style 1 Wooden Box for Flat Sheet and Plate	13
9	Style 1 Plywood or Kraft Overlaid Veneer for Flat Sheet and Plate	13
10	Wirebound Box for Flat Sheet and Plate	14
11	Container Assembly, Fiberboard	15
11A	Double Wall or Triple Wall Fiberboard Box for Flat Sheet and Plate Supported by Skid	16
12	Skid for Long Wide Product	17
13	Beveled Skid Ends	17
14	Skid for Short Bulky Products	17
15	Lengthwise Skid	17
16	Crosswise Skid	17
17	Laminated Lengthwise Skids With Crossmembers	18
18	Lengthwise Skids With Crossmembers	18
19	Machine Wrapped Bundle	18
20	Hand Wrapped Bundle	19
21	Paper Wrapped Coil	19
22	Style 1 Crate	20
23	Style 2 Crate	20
24	Lagged Reel	21
25	Wrapped Reel	21
26	Strapped Bundle for Castings and Forgings	21
27	Coiled Foil Packing	22
28	Skid for Ingot	22
29	Typical Strapped Bundles for Ingot	22

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TABLES

TABLE		PAGE
I	Minimum Requirements for Paper and Barrier Materials	23
II	Species of Woods	24
IIa	Plywood Cross Reference Information	25
III	Boxes for Products Other Than Flat Sheet and Plate (Groups I and II Woods)	26
IIIa	Double Wall and Triple Wall Corrugated Boxes for Extruded and Structural Shapes	27
IIIb	Package Quantity - Flat Plate and Sheet	27
IV	Boxes for Flat Sheet and Plate	28
V	Average Nail Spacing for Boxes of Products Other Than Flat Sheet and Plate	29
VI	Nail Sizes for Boxes of Products Other Than Flat Sheet and Plate	29
VII	Average Nail and Staple Spacing for Boxes of Flat Sheet and Plate	30
VIII	Nail and Staple Sizes for Boxes of Flat Sheet and Plate	31
IX	Strap Sizes for Boxes and Bundles	31
X	Minimum Nail Sizes for Skids	32
XI	Average Nail Spacing and Number of Nails at Intersection of Style 2 Crate Members	32
XII	Strap Sizes for Style 2 Crates	32
XIII	Preservation, Packaging and Packing for Level A	33 - 43
XIV	Minimum Thickness of Flanges for Supporting Coiled Foil	44
XV	Minimum Strap Sizes for Bundles and Skids of Ingot	44
XVI	Preservation and Packaging of Aluminum Sheet and Plate	44
XVII	Preservation and Packaging of Magnesium Sheet and Plate	45
XVIII	Dimensions for Spools of Inert Gas Welding Electrodes	45

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MILITARY STANDARD

ALUMINUM AND MAGNESIUM PRODUCTS
PREPARATION FOR SHIPMENT AND STORAGE

1. INTRODUCTION AND SCOPE

1.1 Scope. This standard describes the minimum requirements for materials, methods, containers and procedures for shipment and storage of aluminum and magnesium products. Although this standard is intended for new procurement, it may also be used by military activities which physically process aluminum and magnesium products.

2. REFERENCED DOCUMENTS

The issues of the following documents in effect on the date of invitation for bids form a part of this standard to the extent specified herein.

SPECIFICATIONS

Federal

FF-N-105	Nails, Wire, Brads and Staples
UU-P-31	Paper, General Specifications and Methods of Testing
NN-P-530	Plywood, Flat Panel
QQ-S-781	Steel Strapping, Flat
QQ-S-790	Steel Strapping, Round (Bare and Zinc Coated)
TT-W-572	Wood-Preservative, Water-Repellent
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-636	Box, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple Wall
PPP-C-96	Cans, Metal, 28 Gage and Lighter
PPP-D-705	Drum: Metal, Shipping, Steel (over 12 and under 55 Gallon)
PPP-D-723	Drums, Fiber
PPP-D-729	Drums: Metal, 55-Gallon (For shipment of non-corrosive material)
PPP-F-320	Fiberboard: Corrugated and Solid Sheet Stock (Container Grade) and Cut Shapes
PPP-P-704	Pails, Shipping Steel (1 through 12 Gallons)
PPP-S-760	Strapping, Non-Metallic, and Connectors
PPP-T-45	Tape, Gummed, Paper Reinforced
PPP-T-60	Tape, Pressure Sensitive Adhesive, Waterproof for Packaging and Sealing
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper (For carton sealing) Water Resistant
PPP-V-205	Veneer, Paper Overlaid (Container Grade)

Military

MIL-M-3171	Magnesium Alloy, Processes for pretreatment and prevention of Corrosion on

STANDARDS

Federal

Federal Standard No. 75	Glossary of Packaging Terms

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads (40" x 48") 4-Way (partial) Pallet Skids, Runners or Pallet-type Base
MIL-STD-731	Quality of Wood Members for Containers and Pallets

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2.1 Other publications. The following documents form a part of this standard. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Non-governmental

CS35 - Hardwood Plywood
PS 1 - Softwood Plywood, Construction and Industrial
CS157 Pine Plywood (Ponderosa Pine, Sugar Pine and Idaho White Pine)

(The above publications are available from Office of Commodity Standards, U. S. Department of Commerce, Washington, D. C. 20234)

Official Classification Committee, Uniform Freight Classification Rules

(The above publication is available from the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N. Y. 10016)

American Trucking Association, National Motor Freight Classification Rules

(The above publication is available from the American Trucking Association, 1616 P Street N. W., Washington, D. C. 20036)

The Aluminum Association Standard, Protective Oil for Aluminum

(The above publication is available from the Aluminum Association, 420 Lexington Avenue, New York, N. Y. 10017)

The Magnesium Association Standard, Protective Oil for Magnesium

(The above publication is available from the Magnesium Association, 1907 Book Building, Detroit, Michigan 48226)

3. DEFINITIONS

3.1 Preservation and packaging. Application or use of adequate protective measures to prevent deterioration including, as applicable, the use of appropriate preservative, protective wrappings, cushioning, interior containers and complete identification marking, up to but not including the exterior shipping container.

3.2 Packing. Application or use of exterior shipping containers, or other shipping media, and assembling of items or packages therein, together with necessary blocking, bracing or cushioning, weather-proofing, exterior strapping, and marking of shipping containers.

3.3 Levels of preservation and packaging

3.3.1 Level A. Military Package. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment, handling, indeterminate storage and world-wide redistribution.

3.3.2 Level C. Minimum military package. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity, for immediate use or controlled humidity storage. This level may conform to suppliers' commercial practice when such meets the requirements of this level.

3.4 Levels of packing

3.4.1 Level A. Military pack. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

3.4.2 Level B. Limited military pack. The degree of packing which will afford adequate protection against damage during multiple shipments. These packs are designed to be shipped and handled under cover and stored in warehouses or other structures having equivalent protection from weather.

3.4.3 Level C. Minimum military pack. The degree of packing which will afford protection against

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damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations and may be the suppliers' commercial practice when such meets the requirements of this level.

3.5 Definition of terms. The following definitions which are in accordance with Federal Standard 75 and any which are in addition thereto are related to the preparation for delivery and storage of aluminum and magnesium products and shall apply:

Additional Cleat (Batten) - A reinforcement member fastened to top, bottom or sides of a container to add strength.

Box - A rigid container having closed faces.

Bundle - A secured unit of two or more articles fastened together.

Cleat - A strip of lumber fastened to a container to add strength.

Container - A receptacle capable of closure.

Corner Protector - Protective material placed under ties to protect edges of a package.

Crate - A rigid container of structural members fastened together. It may be sheathed or unsheathed.

Crossmember - Members placed between deckboards and runners in the construction of skids.

Deckboard - A piece of lumber fastened at right angles to crossmembers, runners, pallet stringers and skids to form a bearing surface.

Edge Protector - Protective material placed between sheared edges of coils of foil and flanges.

Filler - A piece of material placed in a package containing different or irregular sizes for the purpose of squaring out the contents.

Framing Members - Parts which form the fundamental parts of a crate.

Interleaving - The placement of a sheet of protective material between two adjacent pieces of metal.

Loose - A single piece that can be handled individually.

New Weight - The weight of the contents of a package excluding the weight of all packaging and packing material.

Pallet - A low portable platform of wood, metal, fiberboard or combinations thereof to allow the use of fork trucks or slings to facilitate handling, storage and transportation of materials as a unit.

Port Mark - Words, symbols or colors indicating the port of destination.

Separator - Any material used in any direction for dividing and maintaining clearance within a package.

Skid - One of a pair or series of parallel members affixed to the underside of boxes, crates, etc., which serves to strengthen and support the load, to provide for forklift handling and may be longitudinal or transverse.

Tension Tied - Securement applied with mechanical tools.

Wrapped - Unit entirely surrounded with protective material.

4. GENERAL REQUIREMENTS

4.1 Materials, methods and containers. Materials, methods and containers shall conform to the

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requirements of this standard and to the level of protection as specified in the contract or order. Those exceeding the requirements herein may be substituted at suppliers' option when necessary due to availability of materials or equipment.

4.1.1 Materials which are not covered by applicable specifications or which are not specifically described herein shall be of the best quality and suitable for the purpose intended.

4.1.2 Packaging and packing procedures and construction details for containers and systems not covered by specification reference shall be in accordance with 5.5.

4.1.3 The inside dimensions of boxes shall correspond with the length and cross-sectional area of the contents as closely as possible.

5. DETAIL REQUIREMENTS

5.1 Preservation

5.1.1 Cleaning. Surfaces shall be commercially clean, being free of loose mill scale, dirt or corrosion. The presence of residual rolling oil shall not be considered foreign matter.

5.1.2 Preservatives. Oil used for preservation of aluminum products shall conform to the requirements of the Aluminum Association Standard, Protective Oil for Aluminum. Oil used for preservation of magnesium products shall conform to the requirements of the Magnesium Association Standard, Protective Oil for Magnesium. Chrome pickle treatment of magnesium products shall conform to the requirements of MIL-M-3171.

5.1.3 Unless otherwise specified in the contract or order, all surfaces shall be preserved, if required, as specified in Table XIII using materials specified in 5.1 and 5.2.

5.1.4 Magnesium sheet and plate. The preservation and packaging of magnesium sheet and plate when specified in the contract or order shall be in accordance with Table XVII.

5.1.5 Application of Preservative Coating. Compounds shall not be applied to surfaces which show signs of moisture, condensation or frost. Application may be by brushing, dipping, rolling, spraying or flowing onto a surface.

5.2 Preservation and Packaging. Preservation and packaging shall be Level A or C as specified.

5.2.1 Level A. Detailed requirements for preservation and packaging aluminum and magnesium products for Level A are listed alphabetically by product in Table XIII.

5.2.2 Level C. Aluminum and magnesium products shall be preserved and packaged in accordance with the suppliers' standard commercial practice, except that for flat aluminum sheet and plate .090-inches thick and greater, paper interleaving in lieu of oiling (all alloys) shall be provided. (See 3.3.2)

5.3 Packing. In certain cases, a product may be packaged or packed in several types of containers. For such products, if a particular type of container is desired, it shall be specified in the contract or order. Only one type, class or size of material shall be packed in one container except as specified in the contract or order.

5.3.1 Level A. Detailed requirements for packing of aluminum and magnesium products for Level A are listed alphabetically by product in Table XIII. (See 3.4.1)

5.3.2 Level B. Unless otherwise specified, packing shall conform to Level B (see 3.4.2) for domestic shipment.

5.3.2.1 Level C. Aluminum and magnesium products shall be packed in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification Rules or National Motor Freight Classification Rules. (See 3.4.3).

5.4 Materials

5.4.1 Fiberboard. Fiberboard for boxes and liners shall conform to the requirements of PPP-F-320 as applicable.

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5.4.2 Paper and barrier materials. Paper and barrier materials shall be in accordance with Table I.

5.4.3 Wood. The species of wood indicated in Table II and classified into groups as shown shall be used. The groups are set up to include, in any one group, species of woods that have approximately similar characteristics important to box design. These characteristics include density, flexural and compressive strength, stiffness, shock absorption and nail holding power. When any species of wood is specified, any one or more species in the same group may be used. Groups I and II may be interchanged. Groups III and IV may be used in lieu of Groups I and II unless otherwise specified herein. Groups III and IV may be interchanged.

5.4.4 Lumber. All lumber dimensions used in this Standard are commercial sizes except where minimum sizes are specified and shall conform to all requirements of MIL-STD-731.

5.4.5 Plywoods and paper overlaid veneers.

5.4.5.1 Unless otherwise specified, plywood used in the fabrication of domestic type cleated panels shall conform to CS35, Type III, Grade 3-4; PS1, Standard Interior or CS 157, Grade C-D Interior. Plywood used to fabricate overseas type cleated panels shall conform to CS 35, Type I, Grade 3-4; PS 1, Standard Interior with Exterior Glue or CS 157, Grade C-C Exterior. Minimum acceptable grades, types, etc., of plywoods conforming to NN-P-530 and applicable commercial and product standards comparable to types of plywood formerly specified in NN-P-515 are outlined in Table IIa.

5.4.5.2 Unless otherwise specified, paper overlaid veneer used for boxes shall be in accordance with PPP-V-205, Type II for overseas shipment and Type I for domestic shipment.

5.4.6 Nails and staples. Nails and staples shall conform to FF-N-105.

5.4.7 Strapping. Flat steel strapping and round steel strapping shall conform to QQ-S-781, Class A (coated finish) and QQ-S-790, Class A (130,000 - 160,000 psi), Finish 2 (zinc coated) respectively. Equivalent non-metallic strapping conforming to PPP-S-760 may be used when comparable strength requirements are satisfied.

5.4.8 Tapes. Tapes shall conform to the requirements of PPP-T-45, PPP-T-60 or PPP-T-76 as applicable.

5.5 Containers and methods. This paragraph provides general requirements applicable to packaging and packing procedures and construction details for containers and systems not covered by specification reference.

5.5.1 Boxes

5.5.1.1 Boxes for flat sheet and plate. Cleated, Wood (see para 5.4.3), plywood (see para 5.4.5.1) paper overlaid veneer (see para 5.4.5.2), double-wall and triple-wall corrugated fiberboard (see para 5.4.1 and 5.5.1.10) shall be used when fabricating boxes for shipment up to 500 lbs of flat sheet and plate. For cleated panel containers the single piece sides with butt or angle corner joints, the two piece sides cut on an angle with butt or angle corner joints or the multi-piece sides with interlocking corner joints, or the all fiberboard containers with the wrap around and strap closure may be used as applicable. Figures 8 through 11a may be used as a guide for fabrication.

5.5.1.2 Sheet or plate boxes on skids. Unless otherwise specified, a shipment of an individual container to a single destination shall comply with paragraph 5.5.1.10.1 except (a) for a single double-wall or triple-wall fiberboard container paragraphs 5.5.1.10 and 5.5.1.10.1 apply; (b) for a skidded bulk pack, see figures 12 and 14.

5.5.1.3 Two (2) or more containers weighing less than a total of six hundred (600) pounds gross may also utilize the skid system of paragraph 5.5.3 (see figure 11a).

5.5.1.4 Where paragraphs 5.5.1.2 and 5.5.1.3 are not applicable to a contract or order, two (2) to ten (10) boxes, not to exceed six thousand (6000) pounds gross, shall be strapped on skids conforming to figures 15 or 16 with a minimum of one (1) lengthwise and three(3) girthwise 1-1/4-inch by 0.031-inch steel straps.

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5.5.1.5 In addition, when double-wall or triple-wall fiberboard or wirebound boxes are used and unitized as specified in 5.5.1.2 and 5.5.1.3, nominal 1 x 6 battens shall be placed over the top box located under and parallel to the girthwise straps.

5.5.1.6 Boxes for extruded and structural shapes. In addition to wood boxes (see 5.5.1.8) double-wall fiberboard boxes conforming to PPP-B-636 or triple-wall fiberboard boxes conforming to PPP-B-640 may also be used. Table IIIa applies in lieu of the size limitation of the triple-wall box specification.

5.5.1.7 Boxes for products other than extruded or structural shapes, flat sheet and plate. Unless otherwise specified in the contract or order, boxes shall be of wood (see 5.5.1.8) and shall be in accordance with Tables III, V, VI and IX, as applicable, using figures 1 through 7 as a guide.

5.5.1.8 Requirements for wood, plywood and paper-overlaid veneer boxes.

5.5.1.8.1 Ends. Ends 11-1/2 inches or less in height shall be made of one piece, whenever possible. Ends may be laminated from two pieces of equal thickness. Grain of the two laminated pieces shall be at right angles. End thickness as shown by Tables III and IV shall be the combined thickness of the two cross-grained pieces of each end. When the box end exceeds 11-1/2 inches square, styles 2, 4 or 5 ends shall be used.

5.5.1.8.2 End cleats. On styles 2 and 5 boxes, the ends of cleats which are placed across the grain of the end boards shall be one-eighth inch from the inside surface of the top and bottom. The sides, top and bottom shall extend over the cleats and ends. On style 4 box the cleats shall be placed across the grain of the end boards and shall extend within one-eighth inch of the outside surface of the top and bottom. Only the sides shall extend over the cleats.

5.5.1.8.3 Splices. When it is necessary to splice box boards, the boards shall be butted and splice board shall be applied which must extend on each side of the joint at least two times the width of and the same thickness as the box boards. Nails must be clinched.

5.5.1.8.4 Additional cleats. Additional cleats shall be applied to sides, tops or bottoms of all style boxes, except style 1 for flat sheet and plate, when the length of boxes is 60 inches or over and the sides, tops or bottoms consist of more than one piece (see figures 6 and 7). Additional cleats shall be applied to tops and bottoms of style 1 boxes for flat sheet and plate where tops and bottoms consist of more than one piece or are constructed with plywood or paper-overlaid veneer. (See figures 8 and 9).

5.5.1.8.5 Spacing of additional cleats. On all style boxes, except style 1 for flat sheet and plate, additional cleats shall be set in approximately 6 inches from the ends and spaced a maximum of 48 inches apart (see figures 6 and 7). Additional cleats on style 1 boxes for flat sheet and plate shall be placed flush with ends of the tops and bottoms of the boxes and spaced a maximum of 36 inches apart (see figures 8 and 9). Additional lengthwise cleats for style 1 boxes constructed of plywood or paper-overlaid veneer shall be placed flush with the sides of the boxes (see figure 9).

5.5.1.9 Requirements for wirebound boxes of flat sheet and plate. Boxes shall be in accordance with Tables IV, VII, VIII and IX using figure 10 as a guide. End closure of boxes shall be of the twisted type (suitable for hank or power twisters) or the looped type (suitable for Sallee Closer or equivalent).

5.5.1.10 Requirements for double-wall and triple-wall corrugated fiberboard boxes for sheet and plate. Boxes shall be of fiberboard conforming to PPP-F-320, Grade 1100, Class weather-resistant for export shipment and Grade 1100, Class domestic for domestic shipment. Boxes shall be strapped in accordance with Table IX and conform to figure 11.

5.5.1.10.1 Double-wall or triple-wall corrugated fiberboard boxes with contents weighing over 300 pounds shall be supported by wood skids constructed of three 1" x 6" lengthwise deckboards with three 1" x 4" crosswise members (see figure 11a). This requirement applies only to a shipment of an individual container to a single destination (see 5.5.1.5).

5.5.1.11 Palletized shipment. When required by the contract or order, boxes not exceeding 48" long by 40" wide shall be palletized for shipment in accordance with the requirements of MIL-STD-147.

5.5.2 Requirements for all boxes.

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5.5.2.1 Blocking and bracing. Articles which do not completely fill the shipping container shall be blocked, braced, fastened or otherwise secured. Articles having projecting parts that may be broken or may puncture the container shall be rigidly supported, suspended or otherwise protected. Clearance shall be provided between projecting parts and the adjacent inside face of the container. Bracing or blocking shall be prevented from coming in direct contact with any unprotected surface of the item by use of type II paper or as detailed in Table XIII for a specific item.

5.5.2.2 Box skids. On boxes with a gross weight of 350 pounds and over, skids of sufficient thickness to allow for lifting with a fork truck shall be applied to the bottom of the box.

5.5.2.3 Nailing and stapling. Average spacing of nails and staples shall not exceed that shown in Tables V and VIII. When applying cleats, nails shall be driven through the lumber and into the cleat. Nail and staple sizes shall be in accordance with Tables VI and VIII. Nails and staples are not required on double-wall or triple-wall corrugated fiberboard for flat sheet and plate.

5.5.2.4 Strapping. Unless otherwise specified, boxes shall be strapped. Size of strapping shall not be less than as shown in Table IX (nonmetallic strapping).

5.5.2.4.1 Strap spacing. On boxes with additional cleats, girthwise straps shall be placed over the cleats (see figures 6, 7, 8 and 9). On wooden boxes without additional cleats, girthwise straps shall be set in from each end a distance equal to one-sixth the length of the box (see figures 1, 2, 3 and 4) or 6 inches, whichever distance is smaller, and spaced at intervals not exceeding 48 inches (see figure 5). Where it is necessary to splice top, bottom or side boards, girthwise straps placed approximately 6 inches from the splice of the box shall be provided. On wirebound boxes for flat sheet and plate, girthwise straps shall be set in approximately 36 inches from ends and spaced a maximum of 36 inches apart (see figure 10). On double-wall or triple-wall corrugated fiberboard boxes for flat sheet and plate, girthwise straps shall be set in approximately 3 inches from ends and spaced a maximum of 48 inches apart (see figures 11 and 11a).

5.5.3 Skid systems. Skid systems shall be as shown in figures 12 to 18. For long, wide products, skids shown in figures 14, 16 and 28 are used.

5.5.3.1 Skid ends. Skid ends shall be beveled as shown in figure 13.

5.5.3.2 Lengthwise skids. The arrangement shown in figure 15 shall be used for metal from 6 to 14 feet long. For skid systems 60 inches wide and less, two skids shall be used. For heavy rigid metal, two skids may be used for all widths. Skids shall be 2 by 4-inch lumber.

5.5.3.3 Crosswise skids. Figure 16 illustrates the general arrangements for use of skids perpendicular to the deckboards. Skids shall be 2 by 4-inch lumber. Metal less than 6 feet long shall be packed on skid systems using crosswise skids.

5.5.3.4 Lengthwise skids with crossmembers. The arrangements shown in figures 17 and 18 shall be used for metal over 14 feet long. Skids shall be 2 by 6-inch lumber and may be constructed of laminated 1 by 6-inch or spliced 2 by 6-inch lumber. Joints in laminated 1 by 6-inch skids shall occur at crossmembers and shall be staggered as shown in figure 17. Joints in spliced 2 by 6-inch skids shall be reinforced with 2 by 6-inch spliced boards which shall be long enough to contact three crossmembers adjacent to the skid joint as shown in figure 18. Crossmembers shall be 2 by 4-inch lumber, the first crossmember being set in not more than 9 inches from the ends of the skids to the center of the crossmember. Succeeding crossmembers shall be spaced approximately 36 inches apart on center. Interval between the last two crossmembers may vary but shall not exceed approximately 36 inches. Last crossmember is set in not more than 9 inches from ends of skids to center of crossmember. For heavy, rigid metal, spacing of crossmembers may be increased but shall be such as to prevent sagging of metal between crossmembers.

5.5.3.5 Deckboards. Deckboards shall be of 1 by 4-inch lumber spaced to a maximum of 8 inches apart. For heavy, rigid metal, spacing of deckboards may be increased but shall be such as to prevent sagging of metal between deckboards.

5.5.3.5.1 Nailing for deckboards. Nails shall be driven so that the head will not project above the surface of the deckboard. Occasional overdriving of nails will be permitted but no nail shall be overdriven more than 1/2 the thickness of the piece.

5.5.3.6 Sides, ends and tops. Sides, ends and tops, when used, shall be constructed of lumber not less than 1/2 inch thick; minimum 1/8 inch thick hardboard tops may be used in lieu of lumber tops.

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5.5.3.7 End cleats. When end cleats are used they shall be 1 by 4-inch lumber.

5.5.3.8 Top Battens. For lumber tops, top battens shall be 1 by 4-inch lumber. For hardboard tops, top battens shall be minimum 1/8-inch thick by 3-inch wide hardboard. When hardboard battens are used, suitable corner protectors shall be applied under straps which would come in contact with the hardboard.

5.5.3.9 Nailing for skid systems. Minimum size nails shall be as shown in Table X.

5.5.3.10 Strapping for skids. Flat steel strapping shall be minimum 3/4 by 0.023-inch for skids weighing 4,000 pounds net and less, and 1-1/4-inch by 0.031-inch for skids weighing over 4,000 pounds. A minimum of two lengthwise straps shall be used for widths over 60 inches. Crosswise straps shall be placed not more than 9 inches from each end. Except as otherwise specified, intermediate straps shall be applied at approximately 42-inch intervals. Girthwise straps shall pass between the deckboards or crossmembers and skids (see figures 12 and 14).

5.5.4 Bundles. Bundles shall be as shown in figures 19, 20, 21, 26 and 29 and conform to the requirements of Table IX.

5.5.4.1 Wrapping. Unless otherwise specified, coils and long straight lengths shall be either machine wrapped or hand wrapped with materials as specified in Table XIII for specific products.

5.5.4.2 Strapping for bundles and coils. Coils shall be tied in a minimum of 3 places with flat strapping after wrapping (see figure 21). Ties on bundles of straight lengths shall be placed approximately 6 inches from the ends and spaced not more than 48 inches apart (see figures 19 and 20). Size of strapping shall not be less than as shown in Table IX.

5.5.5 Crates construction details. Crates shall be furnished in the following styles:

5.5.5.1 Style 1. Crates without skids for net loads not exceeding 700 pounds and interior dimensions not exceeding 48 inches in length, 36 inches in width and 36 inches in height. (See figure 22).

5.5.5.2 Style 2. Crates with skids for net loads not exceeding 4,000 pounds and interior dimensions not exceeding 40 feet in length, 6 feet in width and 4 feet in height. (See figure 23).

5.5.5.3 Style 1 crates.

5.5.5.3.1 Fabrication. Style 1 crates shall be constructed of 1 by 4-inch lumber framing members, struts and diagonals arranged as shown in figure 22. Diagonals may be placed on the inside or outside.

5.5.5.3.2 Nails. Cement coated or chemically etched 7d nails shall be used. Bright nails may be used in lieu of coated or etched nails for groups III and IV woods. Framing members and struts shall be fastened with a minimum of two nails at each end. Diagonals shall be fastened to other members with a minimum of three nails at each end.

5.5.5.3.3 Strapping. Flat steel strapping 5/8-inch by 0.020-inch or round steel strapping 0.080-inch (14 gage) and equivalent nonmetallic strapping shall be used with comparable strength requirements are satisfied.

5.5.5.3.3.1 Strap spacing. Crates shall be tied with two girthwise straps, one over each vertical framing member. Straps shall be applied perpendicular to the edges of the crate and shall be tensioned so as to sink into the wood at the edges of the crate provided corner protectors are not used.

5.5.5.4 Style 2 crates

5.5.5.4.1 Fabrication. Style 2 crates shall be constructed with skids, deckboards, battens, framing members, ends and sling reinforcement arranged as shown in figure 23.

5.5.5.4.2 Skids. A minimum of two skids shall be used for bases not exceeding 4 feet in width, and a minimum of three skids shall be used for bases 4 feet or greater in width. Skids shall be 2 by 4-inch lumber placed flat.

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5.5.5.4.3 Deckboards. Deckboards shall be of 2 by 6-inch lumber spaced not more than 12 inches from the end of the crate and no more than 60inches apart on centers (see figure 23) and shall be secured to the skids at each intersection. Deckboards shall extend approximately 1 inch beyond each side of the contents of the crate.

5.5.5.4.4 Battens. Battens shall be of 2 by 6-inch lumber grooved to retain the straps placed around them. Top battens shall extend approximately 1 inch beyond each side of the contents of the crate. Side battens shall be approximately 3/8-inch shorter than the height of the contents of the crate. Battens shall be spaced identically to the deckboards.

5.5.5.4.5 Framing members. Framing members shall be 1 by 6-inch lumber.

5.5.5.4.6 Sling reinforcements. Sling reinforcements shall be 2 by 6-inch lumber and shall be positioned on each side at each end between the lower longitudinal framing member and the contents so as to distribute sling pressure over a greater area. These areas shall be designated by marking "SLING HERE". Sling reinforcements shall extend the full distance between two sets of battens.

5.5.5.4.7 Ends. Ends shall be constructed of 1-inch lumber with 1 by 4-inch lumber framing or cleats around the edge of the end. Ends shall have the same dimensions as the width and height of the contents of the crate plus the thickness of the deckboards and battens.

5.5.5.4.8 Nails. Cement coated or chemically etched 7d nails shall be used. Bright nails may be used in lieu of coated or etched nails for groups III and IV woods. Spacing of nails and number of nails to be used in fastening members together shall be as shown in Table XI.

5.5.5.4.9 Strapping. Size of straps shall not be less than as shown in Table XII.

5.5.5.4.9.1 Strap spacing. Straps shall be positioned girthwise over each set of deckboards and battens prior to application of the framing members, ends and sling reinforcements. Straps shall be tensioned so as to sink into the wood at the edges of the battens provided corner protectors are not used, but shall not be tensioned so as to damage the contents of the crates.

5.5.6 Reels. Reels shall be constructed in a manner similar to those shown in figures 24 and 25. Reels with head diameters of 44 inches and less shall be wrapped with V3C solid fiberboard in accordance with PPP-F-320. Fiberboard shall be tied with 0.0915-inch diameter (13 gage) high-tensile round or 1/2-inch by 0.015-inch flat straps or nonmetallic straps when strength requirements are satisfied. Reels with head diameters over 44 inches shall be completely enclosed with wood lagging with lags touching each other. Lags shall be the same thickness as the head but not thicker than 2-inch lumber and shall be nailed to the circumference of the head of the reel extending across edges of both heads. Lags shall be tied with two 0.1620-inch diameter (8 gage) high-tensile round straps, 3/4 x 0.028-inch steel straps or nonmetallic straps of comparable strength. Ties shall be stapled to lags at intervals of approximately 15 inches.

5.5.7 Spools. Spools shall be standard commercial quality and type and size normally used for aluminum and magnesium products.

5.5.8 Steel pails. Steel pails shall conform or be equivalent to the requirements of PPP-P-704.

5.5.9 Steel drums. Steel drums shall conform or be equivalent to the requirements of PPP-D-705.

5.5.10 Metal cans. Metal cans shall conform to the requirements of PPP-C-96, Type V, Class 2.

5.5.11 Fiber drums. Fiber drums shall conform or be equivalent to the requirements of PPP-D-723.

5.5.12 Special containers. Containers other than those listed in this standard which require special design or construction shall be subject to the approval of the procuring activity concerned.

5.5.12.1 Net weight per container. Unless otherwise specified in the contract or order and 5.5.12.2, the net weight per container shall be the maximum acceptable weight specified in Table XIII.

5.5.12.2 When the contract or order specifies a weight that exceeds the maximum permissible weight specified in Table XIII, the procuring activity shall furnish complete packing and container construction details.

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5.5.12.3 When the quantity ordered is less than the minimum weight specified for the product in Table XIII, the quantity or weight shall be specified in the contract or order.

5.5.12.4 Unless otherwise specified in the contract or order, the quantity of flat sheet or plate packed in an individual container shall conform to the unit pack requirements of Table IIIb.

5.6 Markings for shipment. In addition to any special markings required in the contract or order, buncles, containers and loose pieces shall be marked in accordance with MIL-STD-129. When ends and sides of containers are too small to permit marking per MIL-STD-129, the information required shall be marked on top of the containers.

5.6.1 In addition, when the Invitation for Bid or Request for Quotation indicates two units, examples: LBS (pounds), -FT (feet), LBS (pounds) -SH (sheet), shipment and markings shall include the quantity of both units. This marking does not apply for material shipped loose.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

(Copies of this standard for military use may be obtained as indicated in the foreword to the Index of Military Specifications and Standards.)

(Copies of this standard may be obtained for other than official use by individuals, firms and contractors from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20330.)

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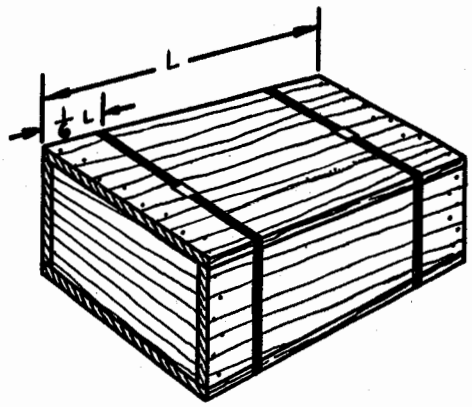


Figure 1.-Style 1 box.

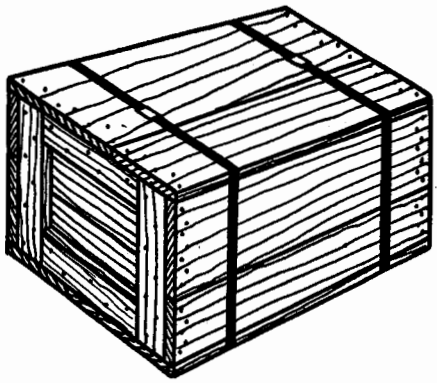


Figure 2.-Style 2 box.

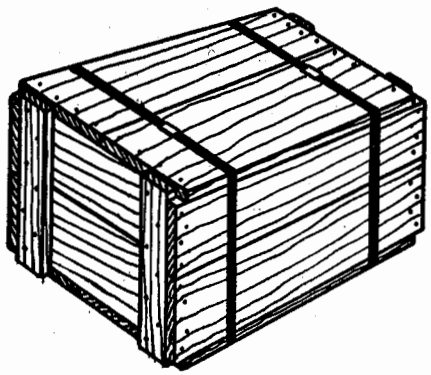


Figure 3.-Style 4 box.

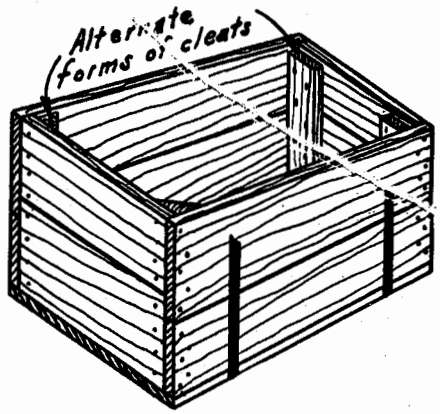


Figure 4.-Style 5 box.

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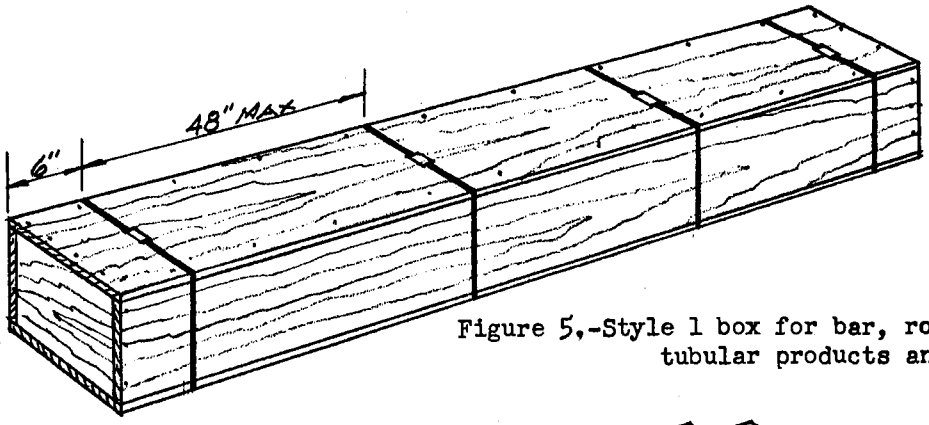


Figure 5.-Style 1 box for bar, rod, shapes, tubular products and wire.

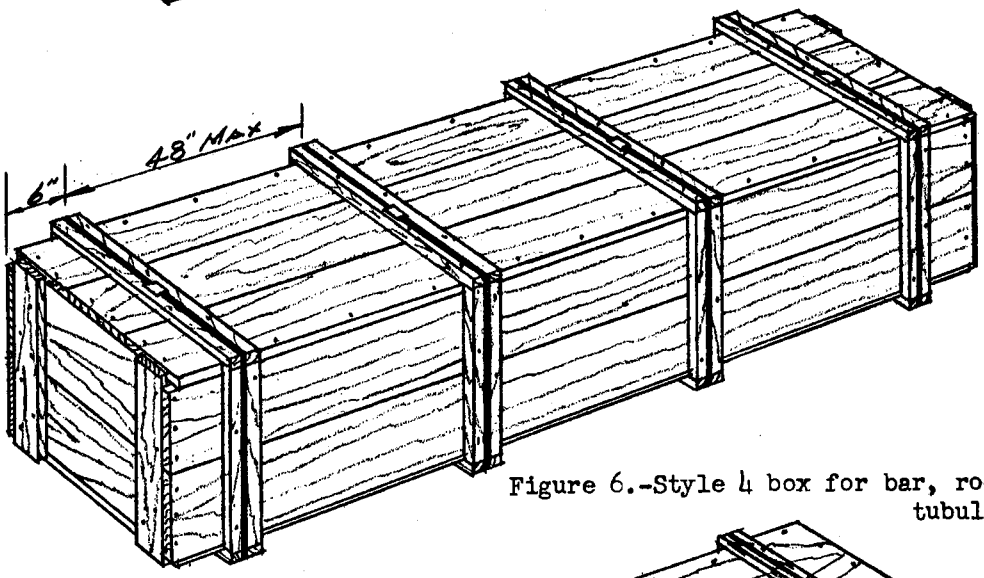


Figure 6.-Style 4 box for bar, rod, shapes, tubular products and wire.

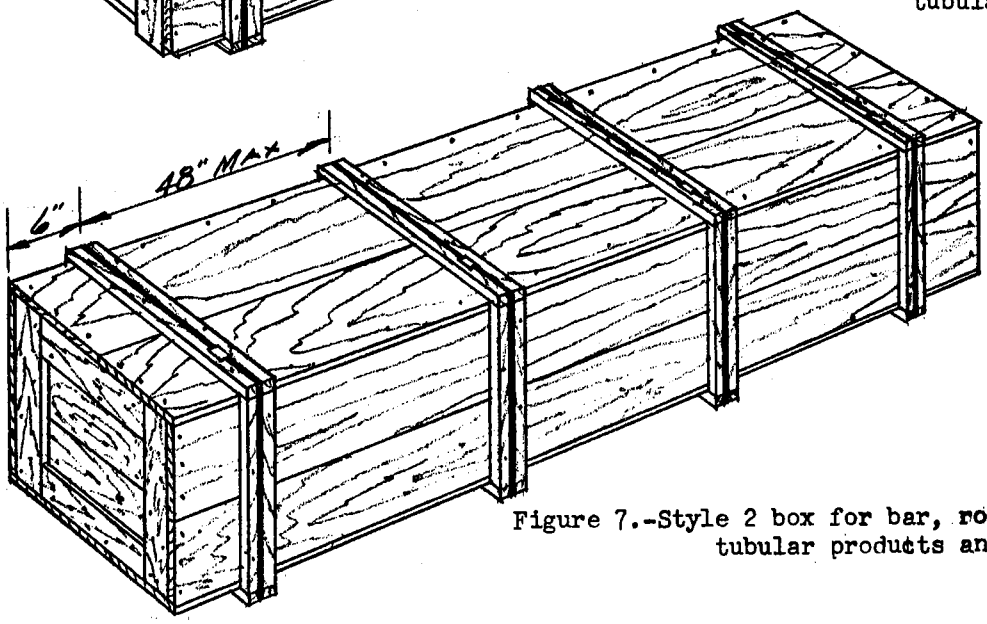


Figure 7.-Style 2 box for bar, rod, shapes, tubular products and wire.

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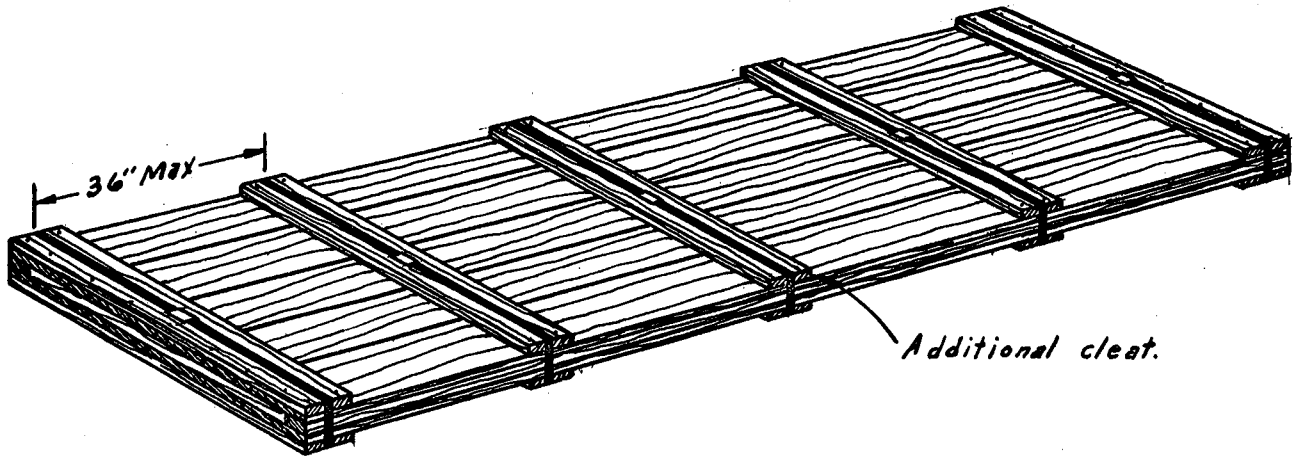


Figure 8.-Style 1 wooden box for flat sheet and plate.

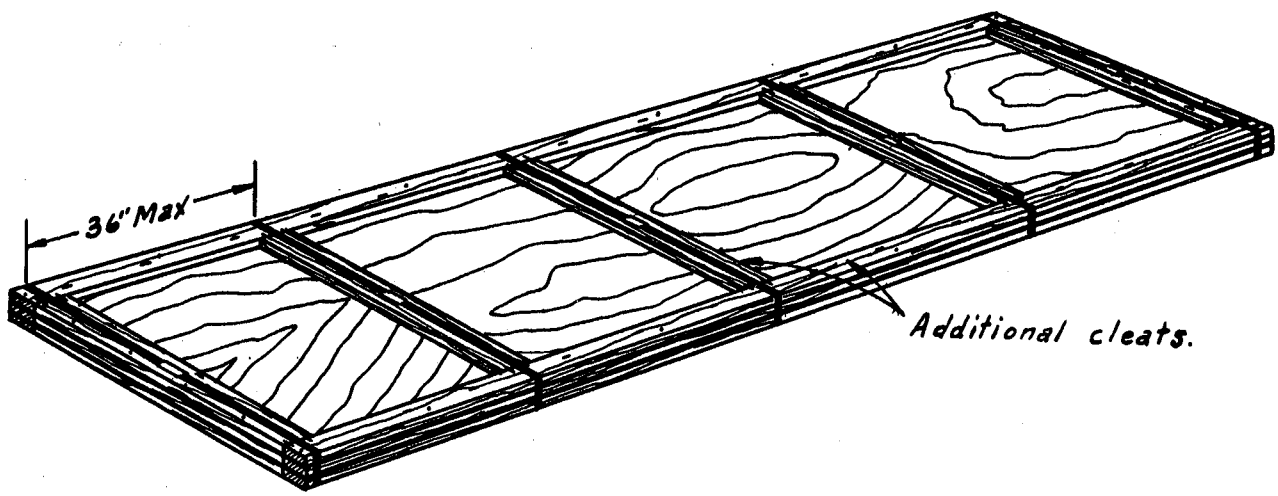


Figure 9.-Style 1 plywood or kraft overlaid veneer box for flat sheet and plate.

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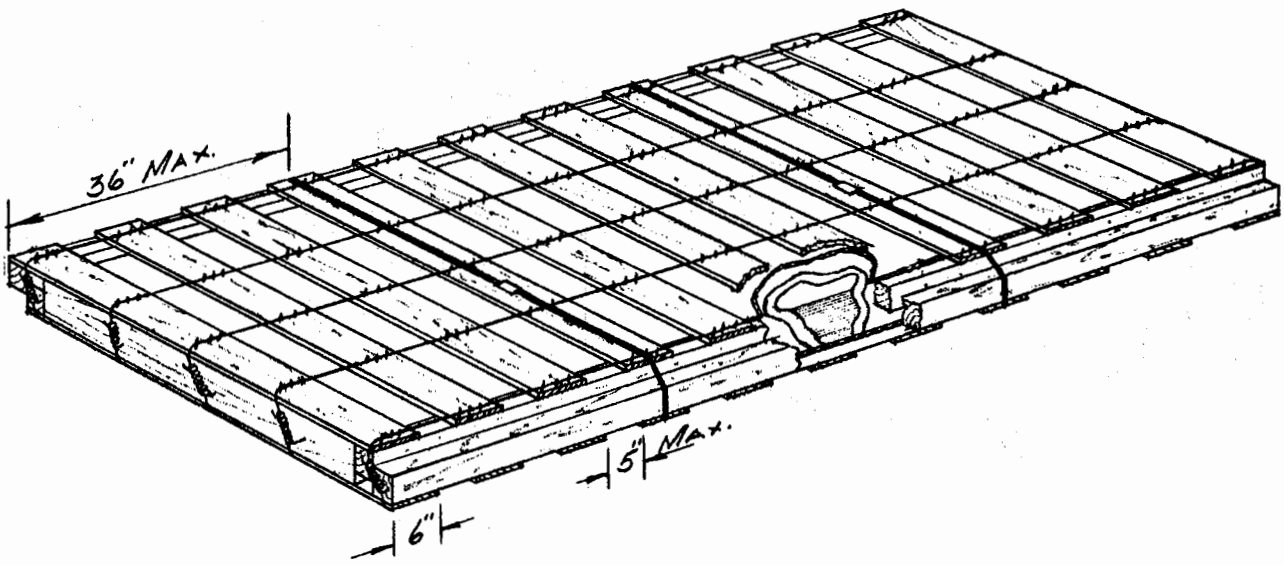


Figure 10 - Wirebound box for flat sheet and plate.

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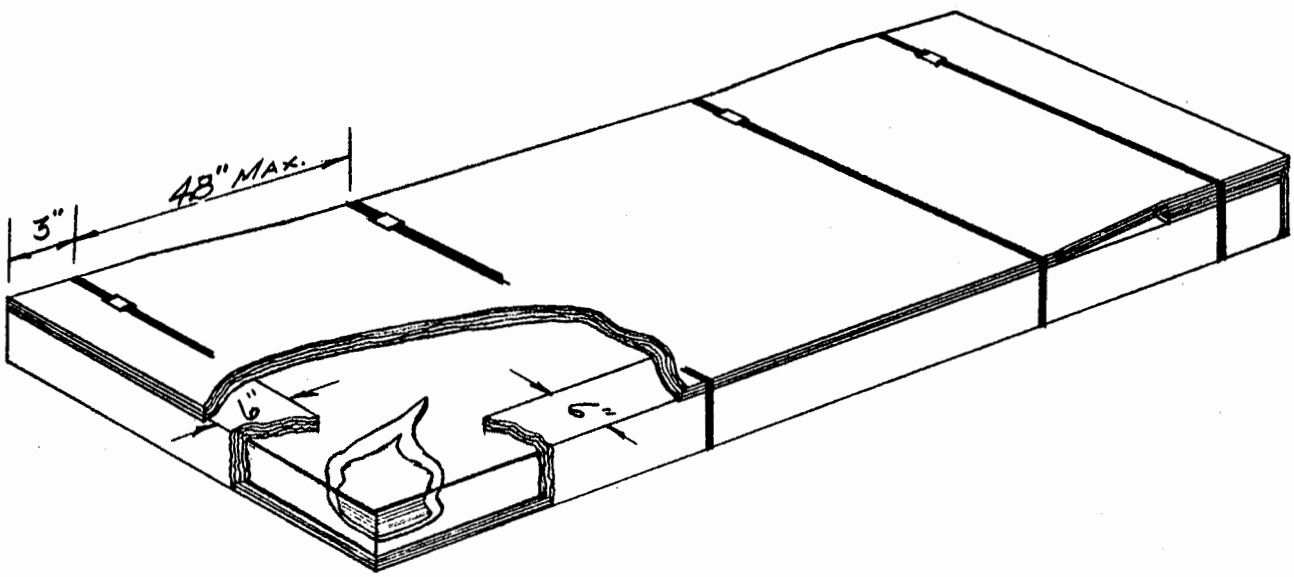


Figure 11 - Triple wall corrugated fibreboard box for flat sheet and plate.

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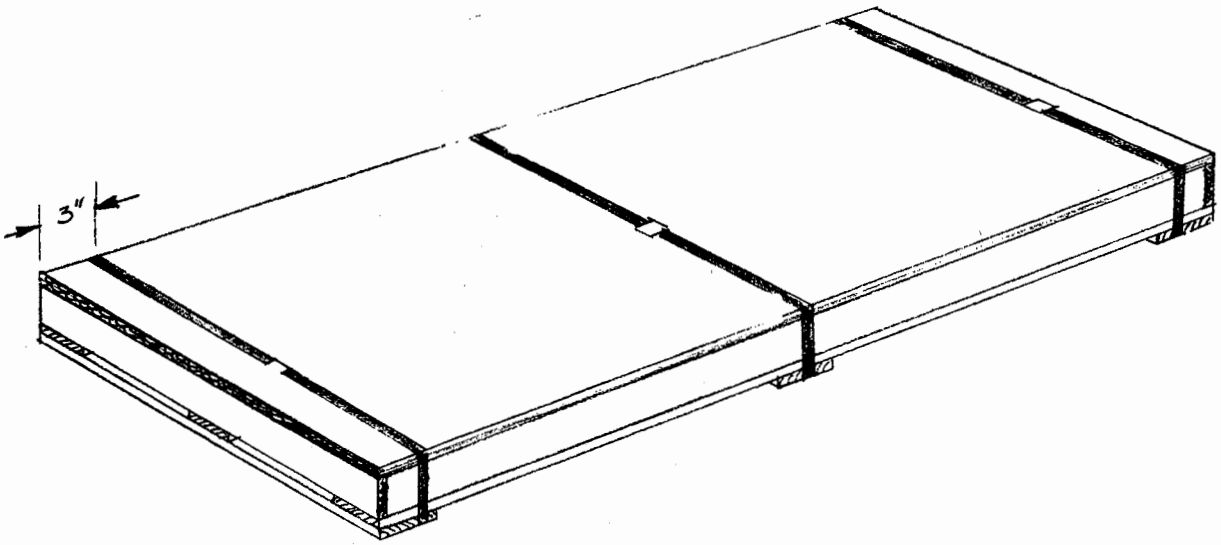


Figure 11A - Triple Wall Corrugated Fiberboard Box for Flat Sheet and Plate Supported by Skid (See paragraph 5.5.1.10.1)

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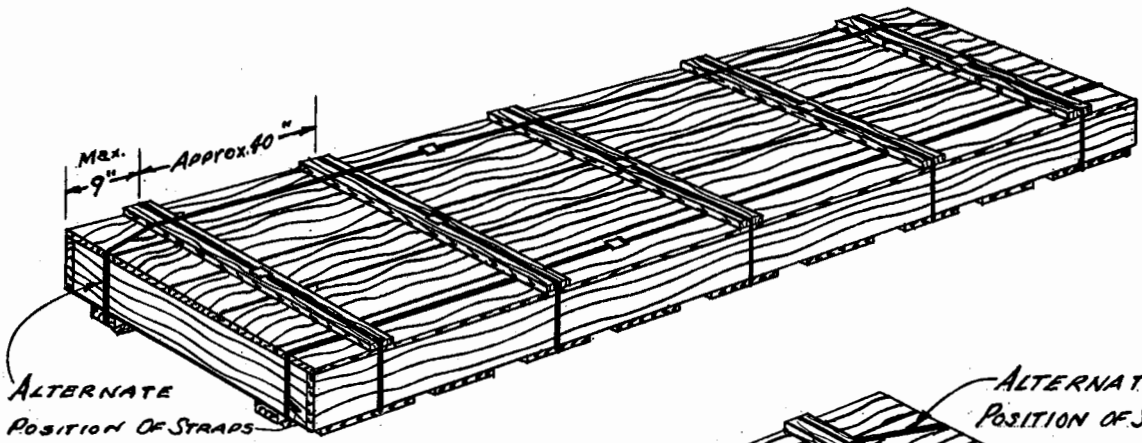


Figure 12.-Skid for long, wide products.



Figure 13.-Beveled ends.

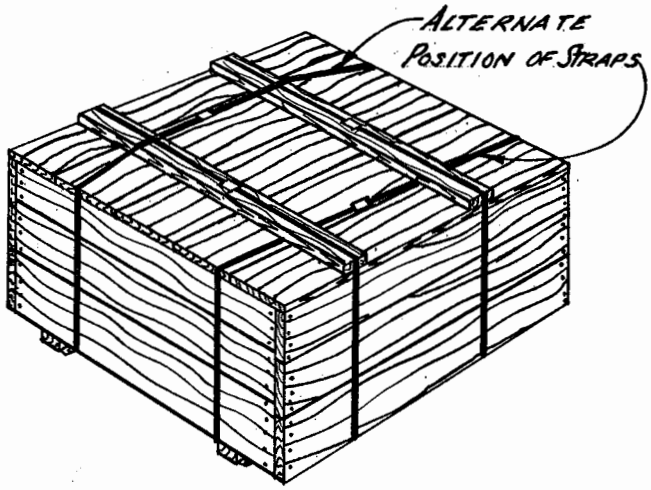


Figure 14.-Skid for short, bulky products.

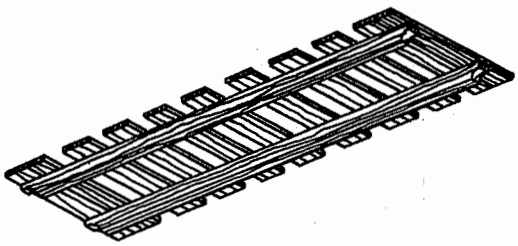


Figure 15.-Lengthwise skid

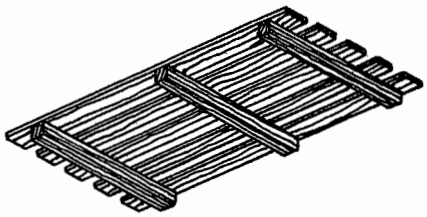


Figure 16.-Crosswise skid

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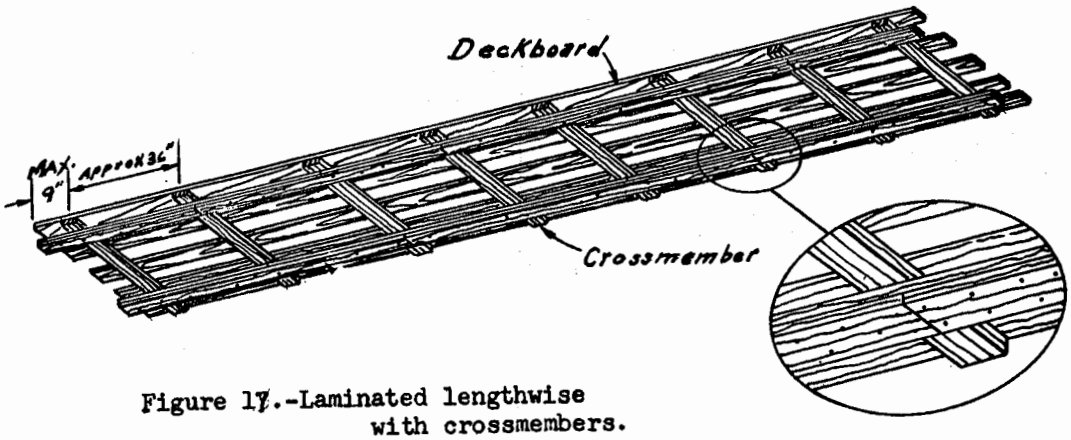


Figure 17.-Laminated lengthwise with crossmembers.

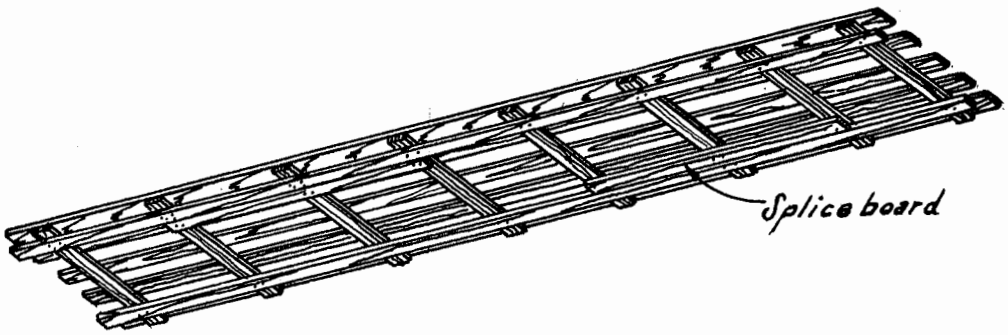


Figure 18.-Lengthwise with crossmembers.

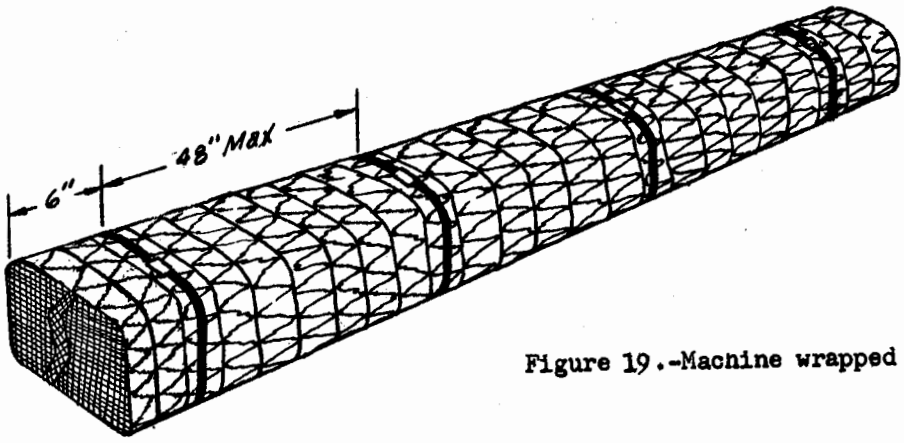


Figure 19.-Machine wrapped bundle.

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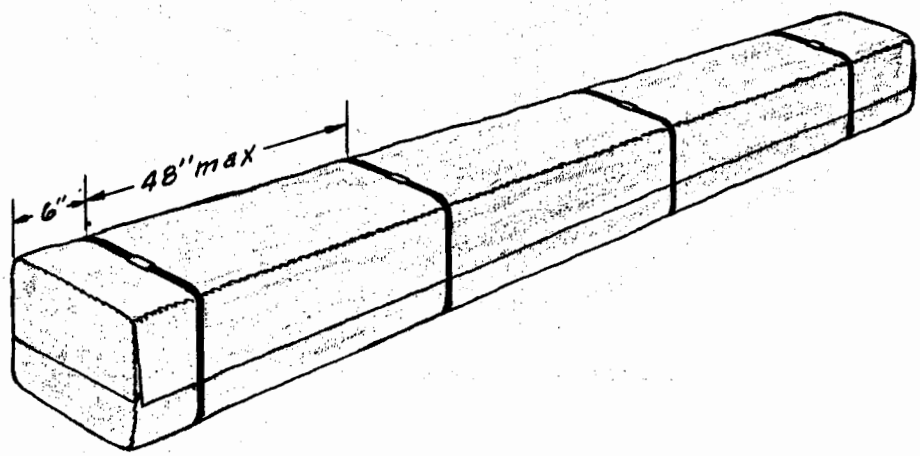


Figure 20.-Hand wrapped bundle.

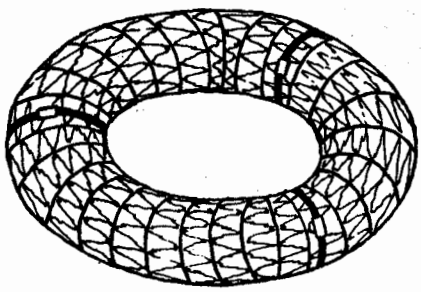


Figure 21.-Paper wrapped coil.

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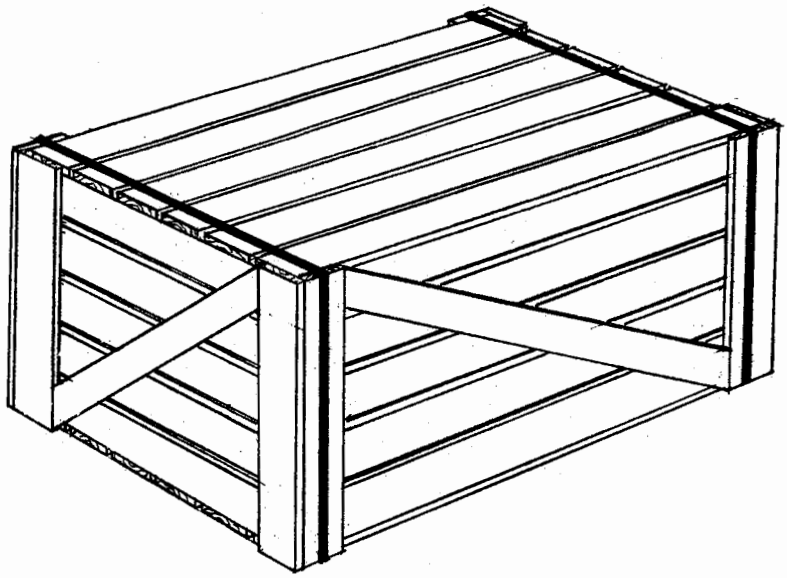


Figure 22. - Style 1 crate

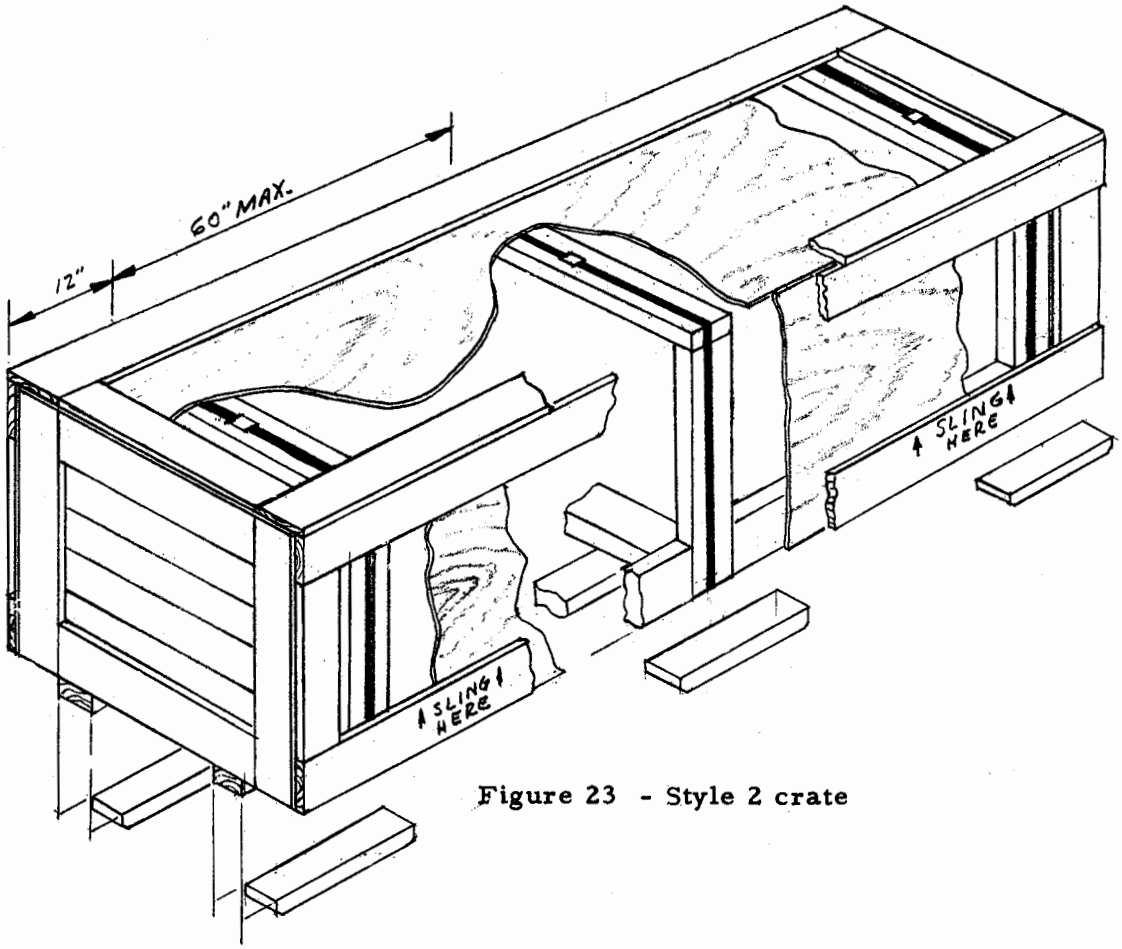


Figure 23 - Style 2 crate

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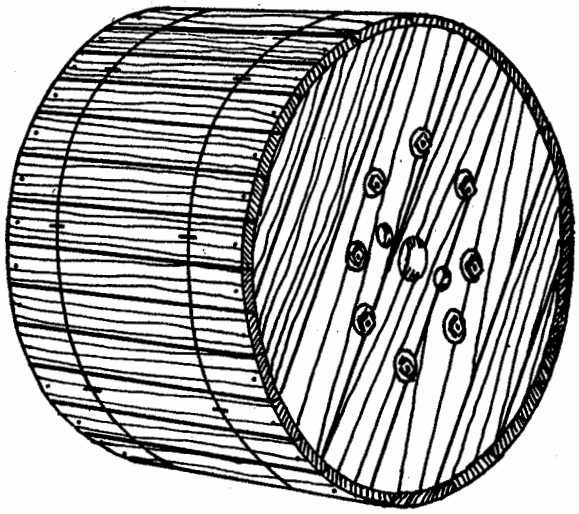


Figure 24.-Lagged reel.

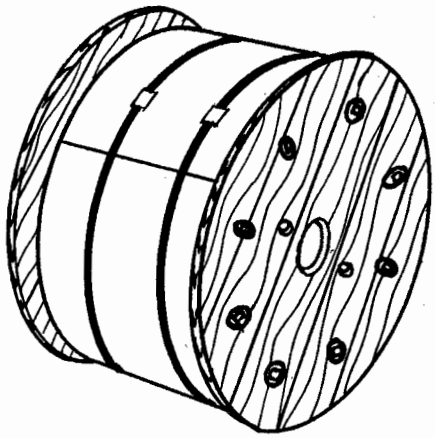


Figure 25.-Wrapped reel.

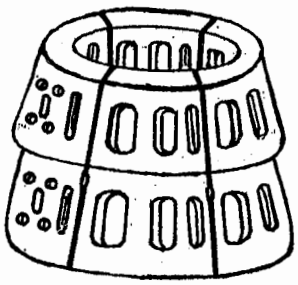


Figure 26.-Strapped bundle for castings and forgings.

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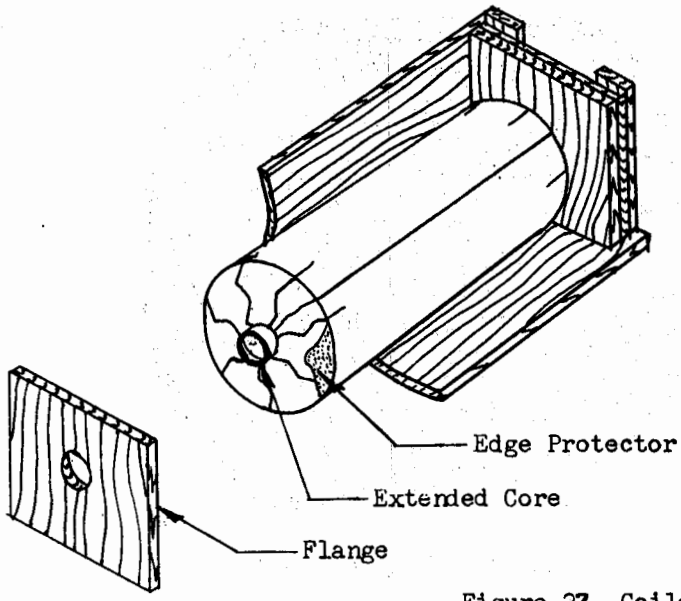


Figure 27.-Coiled foil packing.

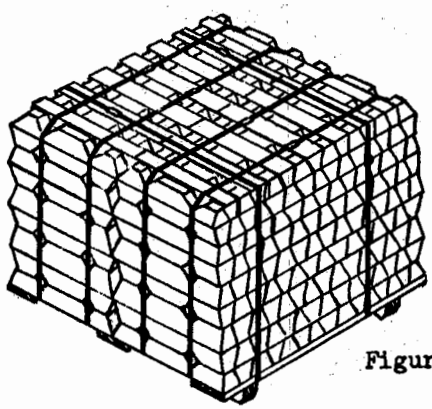


Figure 28.-Skid for ingot.

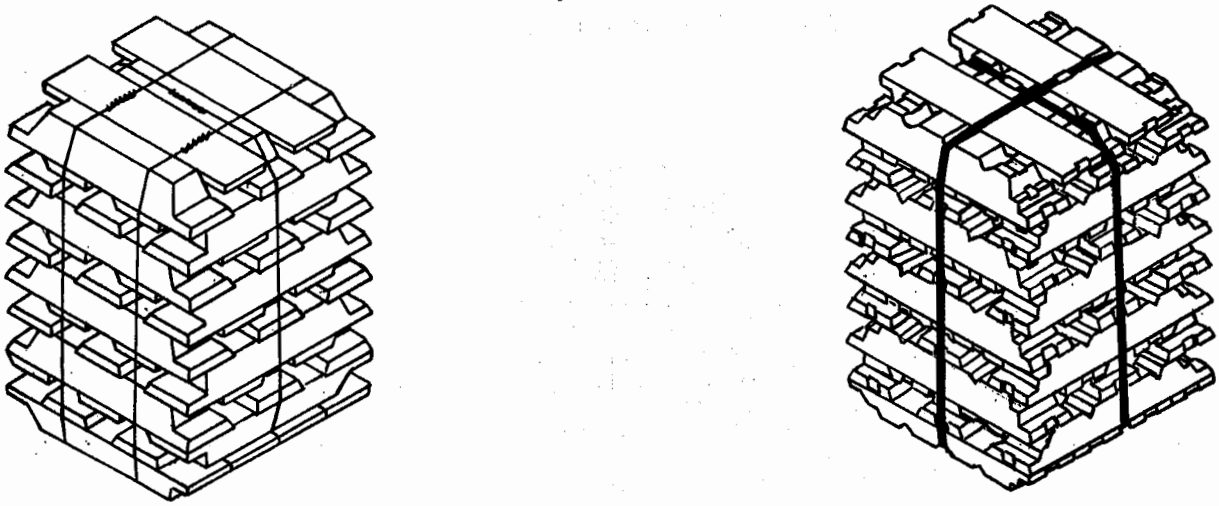


Figure 29.-Typical strapped bundles for ingot.

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TABLE I. Minimum Requirements for Paper and Barrier Materials [1] (See Table XIII)

Types	Combined basis wt., per std. ream 24" x 36"	Tensile strength in weaker direction per inch of width	Stretch in creped direction	Water resistance, dry indicator method
	Pounds	Pounds	Percent	Hours
Type I (interleaving) [2]				
10 lb uncreped	10	1-1/2	- - -	- - -
20 lb uncreped	20	5	- - -	- - -
Type II (general wrapping)				
60 lb. uncreped	60	20	- - -	- - -
Type III (interior wrapping, nonreinforced) 3				
30/12/30 creped	- - -	30	20	1.8
60 lb. uncreped or creped	- - -	15	- - -	2.5
Type IV (exterior and interior wrapping, nonreinforced) 4				
30/30/30 creped	- - -	22	20	2.5
30/30/30 uncreped	- - -	22	- - -	2.5
Type V (exterior and interior wrapping, reinforced) 4				
30/30/30 creped 5	- - -	35 7	- - -	2.5
30/40/30 creped 5	- - -	35 7	- - -	3.0
30/50/30 creped 5	- - -	35 7	- - -	4.0
30/80/30 creped 6	- - -	35	- - -	12.0
30/80/30 uncreped 6	- - -	45	- - -	12.0

1 Any paper or combination of papers or barrier materials which equals or exceeds the minimum requirements shown above may be used in lieu thereof. The mimimum requirements listed in table I are based on the results of tests performed in accordance with the methods outlined in UU-P-31.

2 Type I paper shall be antitarnish, noncorrosive, kraft paper having a pH of 6.5 to 7.5.

3 Type III may be wax or plastic impregnated or laminated with plastic or aluminum foil.

4 Types IV and V asphalt papers shall resist asphalt bleeding for a minimum of 5 hours when suspended in air heated to 170°F.

5 Type V 30/30/30, 30/40/30, and 30/50/30 papers shall have random dispersed reinforcement or a reinforcement spacing not less than 12 threads per foot in the longitudinal direction.

6 Type V 30/80/30 papers shall have random dispersed reinforcement or a reinforcement spacing not less than 12 threads per foot in both directions.

7 Tensile strength shall apply only in the reinforced direction. (Cont'd next page)

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TABLE I (Cont'd)

8 For overseas shipment of sheet and plate aluminum or magnesium, the quantity specified shall be wrapped with a single layer of greaseproof barrier material conforming to Specification MIL-B-121, Type I Grade A. Commercial greaseproof papers equal to or greater than the barrier material specified may be used. Creped material is not acceptable. The barrier material shall be sealed by means of a lockfold or with water resistance tape to prevent loss of the preservation or entrance of free water.

9 Sheet or plate packed as specified shall be wrapped with two layers of 60 pound type III barrier (uncreped) conforming to Table I. The barrier shall have a lock fold or be sealed in such a manner to prevent entrance of free water.

TABLE II. Species of Woods

GROUP I	GROUP II	GROUP III	GROUP IV
Alder, red	Douglas Fir	Ash (cabinet texture)	Ash (tough texture)
Aspen (Popple)	Hemlock	Elm, soft	Beech
Basswood	Larch (Tamarack)	Gum, red or black	Birch
Buckeye	Pine, So. Yellow	Magnolia	Cherry
Butternut		Maple, soft	Elm, hard
Cedar		Sycamore	Hackberry
Chestnut			Hickory
Cottonwood			Locust
Cypress			Maple, hard
Fir (Abies sp)			Oak
Pine (Except So. Yellow)			Pecan
Redwood			
Spruce			
Willow			
Yellow Poplar			

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TABLE IIa

Plywood Cross Reference Information

NN-P-515	CS35	NN-P-530 PS-1	CS157
Type I	Grade 3-4 Type I	Standard Int. with Ext. Glue	Grade C-C Ext.
Type II	Grade 3-4 Type II	Standard Int.	Grade C-D Int.
Type III	Grade 3-4 Type III	Standard Int.	Grade C-D Int.

1. Unless otherwise specified plywood is furnished unsanded. Smooth finish or sanded panels if required must be specified in the contract or order.

2. Plywood outlined in the table is comparable to class 1 formerly specified in NN-P-515.

3. When plywood comparable to class 2 formerly specified in NN-P-515 is required it must be specified that plywood outlined in Table IIa shall be treated with wood preservative; water repellant conforming to TT-W-572.

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TABLE III. Boxes for Products Other Than Flat Sheet and Plate 1 (Groups I and II Woods) 2 (See figs. 1-7)

Weight of contents Pounds	Length of Box Inches	Style of Box	Figure Number	Sides, Top & bottom (minimum thickness) Inch	Ends (min. thick.) Inches	End cleats (minimum thickness & width) Inches	Additional cleats (min. thickness & width) Inches
Less than 350	Less than 60	1	1	1/2	1-1/2	- - -	- - -
Less than 350	60 and over	1	5	5/8	1-1/2	- - -	- - -
Less than 350	Less than 60	2, 4 or 5	2, 3 & 4	1/2	3/4	3/4 x 2-1/2	- - -
Less than 350	60 and over	2 or 4	6 & 7	5/8	3/4	3/4 x 2-1/2	3/4 x 2-1/2
350 to 700	Less than 60	1	1	1/2	1-1/2	- - -	- - -
350 to 700	60 and over	1	5	5/8	1-1/2	- - -	- - -
350 to 700	Less than 60	2, 4 or 5	2, 3 & 4	5/8	3/4	3/4 x 2-1/2	- - -
350 to 700	60 and over	2 or 4	6 & 7	5/8	3/4	3/4 x 2-1/2	3/4 x 2-1/2
700 and over	Less than 60	1	1	5/8	1-1/2	- - -	- - -
700 and over	60 and over	1	5	11/16	1-1/2	- - -	- - -
700 and over	Less than 60	2, 4 or 5	2, 3 & 4	5/8	3/4	3/4 x 2-1/2	- - -
700 and over	60 and over	2 or 3	6 & 7	11/16	3/4	3/4 x 2-1/2	3/4 x 2-1/2

1 Includes circles less than 36 inches diameter. For circles 36 inches diameter and over see table IV.

2 When groups III and IV are used in lieu of groups I and II, lumber man be one-sixteenth inch less in thickness than that specified for groups I and II.

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TABLE IIIa. Double wall and triple wall corrugated boxes for extruded and structural shapes

Weight of Contents Max. Pounds	Style of Box Ref PPP-B-636 & PPP-B-640	Inside Dimensions Max. Inches L x W x D
300	A - B - C	192 x 12 x 12
300	A - B - C	144 x 18 x 18
300	A - B - C	144 x 24 x 24

MIL-B-43291(GS) Boxes, Fiberboard, Corrugated, Double-Wall Weather Resistant

* TABLE IIIb. Package Quantity - Flat Plate & Sheet (See Note 4.)

* Dimensions in Inches (See Note 5) Thickness	Width	Length	Max. Quantity Per Container
Less than .006	Any	Any	Package and pack as specified for foil
.006 thru .249	Less than 36	Less than 144	25 or less
.250 and over	Less than 36	Less than 144	10 or less
.006 thru .249	36 thru 60	to 144 incl.	5 (see note 1 & 2)
.250 and over	36 thru 60	to 144 incl.	2 (see note 1 & 2)
.006 and over	Over 60	Any	- (see note 3)
.006 and over	Any	Over 144	- (see note 3)

NOTES:

1. The final box for any one item may contain less than twice the specified number of pieces per box - provided the final box does not exceed the specified standard net weight for that box.

2. If the net weight of 5 sheets of 2 plates exceeds the weight limitations of Table XIII, either a lesser quantity per box, or skid systems (Fig. 12 and 14) will be utilized.

3. Package quantity shall be governed by the net weight limitations under Table XIII, for the type container used.

4. Table IIIb does not apply for material destined to Navay Shipyards or for contracts and orders specifying special support programs, i.e., FRAM. (Fleet Rehabilitation and Maintenance.)

5. Industry standard width and length rolling or cutting tolerances shall not be misconstrued to place the required standard (procured size) into the next higher or lower table requirement.

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TABLE IV. Boxes for Flat Sheet and Plate
(see figs. 8 - 11)

Style of Box	Figure number	Depth of box Inches	Sides (minimum thickness) 1	Top and bottom (minimum thickness) 2	Ends (minimum thickness) 1 Inches	End cleats (minimum thickness and width) Inches	Additional cleats (top and bottom) (min. thickness & width) Inches
Wood, plywood or paper-overlaid veneer (groups I and II woods) 3							
1	8 & 9	Less than 2-1/2	1-1/2	1/2	1-1/2	---	1/2 x 2-1/2
1	8 & 9	2-1/2 to 8	5/8	1/2	1-1/2	---	1/2 x 2-1/2
2 or 4	2, 3, 6 & 7	8 and over	5/8	1/2	3/4	3/4 x 2-1/2	3/4 x 2-1/2
Wirebound (group III wood) 4				5 6			
---	10	1/2 to 3/4	7/8 thick x 1/2 high	5/16 x 6	7/8 thick x 1/2 high	---	---
---	10	3/4 to 1	7/8 thick x 7/8 high	5 6 5/16 x 6	7/8 thick x 7/8 high	---	---
Double wall and triple wall corrugated fiberboard							
---	11 plus 11A	---	---	---	---	---	---

1 Thickness is that dimension measured from inside to outside of box.

2 Three-twentieth inch minimum thickness three-ply plywood or one-eighth inch thick core paper-overlaid veneer may be used in lieu of one-half inch thick lumber tops and bottoms.

3 When groups III and IV are used in lieu of groups I and II, lumber may be one-sixteenth inch less in thickness than that specified for groups I and II.

4 Top and bottom of wirebound boxes are lined with 125 pound test double-faced corrugated fiberboard.

5 When group IV is used in lieu of group III, lumber may be one-sixteenth inch less in thickness.

6 Top and bottom boards shall be evenly spaced with not more than 5 inches space between adjacent boards.

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TABLE V. Average Nail Spacing for Boxes of Products other than Flat Sheet and Plate 1 (See figs. 1 - 7)

Parts nailed	Nail spacing Inches
Sides, top and bottom to additional cleats	4
End cleats to ends	2
Sides, top and bottom to ends and end cleats	2
Top and bottom to sides	10

1 Includes circles less than 36 inches diameter. For circles 36 inches diameter and over see Table VII.

TABLE VI. Nail Sizes for Boxes of Products Other Than Flat Sheet and Plate 2 (see figs. 1 - 7)

Parts nailed	Thickness of parts holding nail heads Inch	Nail Size
Sides, top and bottom to additional cleats	1/2	4d
	5/8	5d
	11/16	5d
End cleats to ends	3/4	5d
Sides, top and bottom to ends and end cleats	1/2	6d
	5/8	7d
	11/16	8d
Top and bottom to sides 3	11/16	7d

2 Includes circles less than 36" diameter. For circles 36" and over see table VIII.

3 Top and bottom are not nailed to sides less than 11/16" thick.

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TABLE VII. Average Nail and Staple Spacing for Boxes of Flat Sheet and Plate (see figs. 8-11)

PARTS FASTENED	SPACING INCHES
Top and bottom to additional cleats at ends of wood box	4
Top and bottom to additional crosswise cleats on plywood and paper-overlaid veneer boxes	12
End cleats to end	2
Top and bottom to additional cleats at side of box	9
Top and bottom to ends	6
Top and bottom to sides	
Wood	2 nails at ends of additional cleats
Plywood	18
Paper-overlaid veneer	18
Wirebound, Wires to top and bottom boards	3 per board
Sides to ends and end cleats. 1	2
Triple wall corrugated Fiberboard	None required

1 Sides less than 1-1/2" high are not nailed to ends.

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TABLE VIII. Nail and Staple Sizes for Boxes of Flat Sheet and Plate (see figs. 8-11)

Parts nailed or stapled	Thickness of parts holding nail head or staple crown	Nail or staple size
	Inches	
Top and bottom to additional cleats		
Wood	1/2	4d
Plywood	3/20	1
Paper-overlaid veneer	1/8 core	1
End cleats to ends [2]	3/4	5d
Sides to ends and end cleats	1-1/2	10d
	5/8	7d
Top and bottom to sides and ends		
Wood, plywood or paper-overlaid veneer	1/2	5d to 12d [3]
Wirebound		
Wires to top and bottom boards	1/4	0.0475" (18 gage) x 7/16" [4]
Wires through top and bottom boards into sides	1/2 sides	0.0625"(16 gage) x 3/4"
	7/8 sides	0.0625"(16 gage) x 1-1/8"
Double wall and triple wall corrugated fiberboard	---	

1 3d nails or minimum 0.0625-inch diameter (16 gage) staples having minimum 1/2-inch crown and legs whose length equals sum of thickness of plywood or paper-overlaid veneer and cleat plus 1/8-inch.

2 Sides less than 1-1/2 inches high are not nailed to ends.

3 Size of nail varies with height of sides.

4 Staples are clinched to produce a smooth surface on inside of box.

TABLE IX. Strap Sizes for Boxes and Bundles (see 5.4.7) [6] (see figs 1-11, 19-21 and 26)

Weight of contents	Flat steel strapping	Round steel strapping	Binding wires for wirebound boxes [5]
Pounds	Inch	Inch	Inch
Less than 700	5/8 x 0.020	0.0800 (14 gage)	0.0800 (14 gage)
700 and over	3/4 x 0.023	0.1055 (12 gage)	0.0800 (14 gage)

5 Black annealed steel wire:
Five wires for boxes less than 54-inches wide.
Six wires for boxes 54-inches wide and over.

6 PPP-S-760a Strapping, Nonmetallic, (and Connectors).

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TABLE X. Minimum Nail Sizes for Skids (See figs 12-18)

Parts Nailed	Nail Size				
	Cement coated sinker			Helical threaded or annular grooved	
	Penny	Length Inches	Diameter Inch	Length Inches	Diameter Inch
Crossmembers to runners	16	3-1/4	0.1483	3-1/4	0.1483
Deckboards to runners or crossmembers	10	2-7/8	0.1205	2-1/2	0.1200

TABLE XI. Average Nail Spacing and Number of Nails at Intersection of Style 2 Crate Members (see fig. 23)

Parts Nailed	Spacing Inches	Number of Nails at Each Intersection
Deckboards to runners	---	3
Long edges of adjacent framing members	6	---
Ends of framing members to ends	---	3
Framing members to battens	---	4
Framing members to sling reinforcement	6	---
End framing members to end sheathing	6	2

TABLE XII. Strap Sizes for Style 2 Crates (see fig 23)

Weight of Contents Pounds	Flat Steel Strapping Inch	Round Steel Strapping Inch
Less than 700	5/8 x 0.020	0.080 (14 gage)
700 to 2,000	3/4 x 0.023	0.105 (12 gage)
2,000 and over	3/4 x 0.031	0.162 (8 gage)

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TABLE XIII. Preservation, Packaging and Packing for Level A (For level C, see 5.2.2)

Product	Preservation Level A (See 5.1)	Packaging Level A (See 5.2)	Packing Level A (see 5.3) (For Battiers see Table I) [1]	Standard Net Weight Per Container (See 5.5.12.1)
Bar, Rod and Wire (cold-finished, drawn, extruded and rolled)				Pounds
Coiled, bare	A1-Oiled Mg-Chrome pickled	None	Wrapped coils (see fig. 21). Wrapping shall be one layer of 30/12/30, type II barrier and one layer of 30/30/30, type V barrier.	55 or 110 to 120
Coiled, covered	None	None	Wrapped coils (see fig. 21). Wrapping shall be two layers of 30/30/30, type V barrier.	55 or 110 to 120
Spooled	None	Five, ten, fifteen, twenty or thirty pounds per spool	Wooden boxes (see figs. 1-4) Boxes shall be lined with two layers of 60 pound, type III barrier.	150 to 300
Straight lengths	A1-Oiled Mg-Chrome pickled	None	Wooden boxes (see figs. 5-7) Boxes shall be lined with two layers of 60 pound, type III barrier.	300 to 2,000
	A1-Oiled Mg-Chrome pickled	None	Skids (see figs. 12-18) Skids shall be lined with one layer of 60 pound, type III and one layer of 30/80/30 type V barrier.	2,000 to 6,000
	A1-Oiled Mg-Chrome pickled	None	Double wall and triple wall (Corrugated Fiberboard boxes (see paragraph 5.5.1.6 and table IIIa)	To 300
Blooms and Billets	None	None	See ingot.	---

[1] Where multiple layers of barriers are required, a single layer of barrier material which equals or exceeds the protective qualities of the multiple layers may be used in lieu thereof.

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TABLE XIII. Preservation, Packaging and Packing for Level A (For level C, see 5.2.2)

Product	Preservation Level A (See 5.1)	Packaging Level A (See 5.2)	Packing Level A (see 5.3) (For Barriers see table I) [1]	Standard Net Weight Per Container (See 5.5.12.1) Pounds
Bus Conductors (cold-finished, drawn, extruded and rolled)	A1-Oiled Chrome pickled	None	Wooden boxes (see figs. 5-7). Boxes shall be lined with two layers of 60 pound, type III barrier.	300 to 2,000
	A1-Oiled Mg-Chrome pickled	None	Skids (see figs. 12-18). Skids shall be lined with one layer of 60 pound, type III and one layer of 30/80/30, type V barrier.	2,000 to 6,000
Cable (bare and covered)				
Size 1/0 and smaller	None	None	Wrapped coils (see fib. 21). Wrapping shall be two layers of 30/30/30, type V barrier.	Bare cable - 150 to 250 Covered cable - 50 to 200
	None	None	Reels (see figs. 24-25).	Bare cable - 700 to 1,250 Covered cable - 125 to 1,000
Size larger than 1/0	None	None	Reels (see figs. 24-25).	Bare cable - 700 to 9,600 Covered cable - 350 to 1,300
Castings	None	None	Wooden boxes (see figs. 1-4). Boxes shall be lined with two layers of 60 pound, type III barrier.	100 to 700
	None	None	Crates (see fig. 22). Crates shall be lined with two layers of 60 pound, type III barrier.	100 to 700
	None	None	Bare bundles (see fig. 26)	100 to 400
	None	None	Loose	---

[1] Where multiple layers of barriers are required, a single layer of barrier material which equals or exceeds the protective qualities of the multiple layers may be used in lieu thereof.

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TABLE XIII. Preservation, Packaging and Packing for Level A (For level C, see 5.2.2)

Product	Preservation Level A (See 5.1)	Packaging Level A (See 5.2)	Packing Level A (see 5.3) (For Barriers see table I)[1]	Standard Net Weight Per Container (See 5.5.12.1)
				Pounds
Conduit	None	External threads shall be covered with suitable thread protectors	Wooden Boxes (see figs 5-7). Boxes shall be lined with two layers of 60 pound, type III barrier.	300 to 2,000
Extruded shapes (Material less than 1-pound per linear foot)	A1-Oiled Mg-Chrome pickled	None	Wooden Boxes (see figs. 5-7). Boxes shall be lined with two layers of 60 pound, type III barrier.	300 to 2,000
	A1-Oiled Mg-Chrome pickled	None	Double wall and triple wall corrugated fiberboard boxes (see par 5.5.1.6) and table IIIa.	To 300
Fittings (pipe and rigid conduit)	A1-Oiled Mg-Chrome pickled	External threads shall be covered with suitable thread protectors	Wooden boxes or crates dependent upon size of fittings (see figs. 1-4 and 22). Boxes or crates shall be lined with two layers of 60 pound, type III barrier.	50 to 100
Foil[2]	None	Foil shall be wound on aluminum, fiber or steel cores. End of coil shall be secured with pressure sensitive tape. Sheared edges shall be protected from flanges and adjacent coils with suitable edge protectors. Each coil on extended core or wood dowel or several narrow coils on one dowel shall be wrapped with minimum .001" thick aluminum foil. Foil wrapping shall be a conformable wrap completely enclosing the coil or coils and edge protectors on each core or dowel. (See fig. 27)	Wooden boxes (see figs. 1-4). Coils shall be suspended by extended cores or wood dowels thru the core. Core extension or dowel shall be inserted in wood flanges so that periphery of coil does not contact inner surface of box (see fig. 27). Minimum flange thickness shall be as specified in table XIV.	150 to 500
Coiled	None	Same as above	Double wall and triple wall corrugated fiberboard boxes (see 5.5.1.6)	To 300

1 Where multiple layers of barriers are required, single layer of barrier material which equals or exceeds the protective qualities of the multiple layers may be used in lieu thereof.

2 Foil is sheet metal less than 0.006 inch thick.

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TABLE XIII. Preservation, Packaging and Packing for Level A (For level C, see 5.2.2)

Product	Preservation Level A (See 5.1)	Packaging Level A (See 5.2)	Packing Level A (See 5.3) [1] (For Barriers see table I)	Standard Net Weight Per Container (See 5.5.12.1)
				Pounds
Flat	None	None	Wooden boxes (see figs. 1-4). Boxes shall be lined with two layers of 60 pound, type III barrier.	150 to 500
Flat	None	None	Double wall and triple wall corrugated fiberboard boxes (see 5.5.1.6).	To 300
Forgings	None	See Castings	See Castings	---
Forging Stock	A1-Oiled Mg-Chrome pickled	See Bar	See Bar	---
Ingot				
500 pounds per piece and over	None	None	Loose	---
30 pounds to but not including 500 pounds per piece	None	None	Bare bundles (see fig. 29). Size of bundle straps shall be as shown in table XI. A minimum of two straps shall be used per bundle.	2,000 to 2,700
Less than 30 pounds per piece	None	None	Skids without top, sides or ends (see fig. 28). Size of skid straps shall be as shown in table XV.	2,000 to 2,500
Grained and granulated ingot	None	PP-B-566 Boxes, Folding Paper Board	Wooden boxes (see figs. 1-4). Boxes shall be lined with two layers of 60 pound, type III barrier.	500
	None	None	PPP-P-704 steel pails (see 5.5.8). Pails shall not be overpacked.	70
	None	None	PPP-D-705 steel drums (see 5.5.9). Drums shall not be overpacked.	450 to 600

[1] Where multiple layers of barriers are required, a single layer of barrier material which equals or exceeds the protective qualities of the multiple layers may be used in lieu thereof.

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TABLE XIII. Preservation, Packaging and Packing for Level A (For level C, see 5.2.2)

Product	Preservation Level A (See 5.1)	Packaging Level A (See 5.2)	Packing Level A (See 5.3) (For Barriers see table I)	Standard Net Weight Per Container (See 5.5.12.1)
				Pounds
	None	None	PPP-D-723 drums fiber (see 5.5.11). Drums shall not be overpacked.	To 500
Nails	None	None	Wooden boxes (see figs. 1-4). Boxes shall be lined with two layers of 60 pound, type III barrier.	200
Paste and Powder	None	One, two or ten pounds per friction-top can per PPP-C-96 type V Class 2	Wooden boxes (see figs. 1-4). Boxes shall be lined with one layer of 60 pound, type III barrier.	50
	None	None	PPP-D-705 steel drums (see 5.5.9). Drums shall not be overpacked.	50 to 550
			PPP-D-723 fiber drums constructed and packaged in conformance with MIL-A-512A. Drums shall not be overpacked.	
Rivets, Screw Machine Products and Washers				
Rivets and screw machine products	A1-Oiled Mg-Chrome pickled	One of five pounds per full telescoping, paperboard box per PPP-B-566. Paperboard shall be bending boxboard type, with a minimum thickness of 0.020 inch. Boxes shall be sealed with one strip of 2 inch wide gummed paper tape PPP-T-45 running lengthwise over bottom and extending up the ends.	Wooden boxes (see figs. 1-4). Boxes shall be lined with one layer of 30/30/30, type IV barrier.	400

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TABLE XIII. (Cont'd)

Product	Preservation Level A (See 5.1)	Packaging Level A (See 5.2)	Packing Level A (See 5.3) (For Barriers see table I)[1]	Standard Net Weight Per Container (See 5.5.12.1)
				Pounds
Washers	None	100 pieces per number 5-1/2, substance 28, kraft paper envelopes.	Wooden boxes (see figs. 1-4). Boxes shall be lined with one layer of 30/30/30, type IV barrier.	400
Screw Machine Stock	A1-Oiled Mg-Chrome pickled	See bar	See bar	---

1 Where multiple layers of barriers are required, a single layer of barrier material which equals or exceeds the protective qualities of the multiple layers may be used in lieu thereof.

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TABLE XIII. Preservation, Packaging and Packing for Level A (For level C, see 5.2.2)

Product	Preservation Level A (See 5.1)	Packaging Level A (See 5.3)	Packing Level A (See 5.3) [1] (For Barriers see Table I)	Standard Net Weight Per Container (See 5.5.12.1)
				Pounds
Sheet and plate[3] Flat and tapered sheet and plate, perforated sheet, and plate circles	A1-See Table XVI Mg-See Table XVII	A1-See Table XVI Mg-See Table XVII	Skids (see figs. 12-18. Skids shall be lined with one layer of 60 pound, type III and two layers of 20/80/30 type V barriers.	Sheet: 1,000 to 6,000 Plate: 1,000 to 6,000
	A1-See Table XVI Mg-See Table XVII	A1-See Table XVI Mg-See Table XVII	Wood, plywood or paper-overlaid veneer boxes (see figs. 8-9). Boxes shall be lined with two layers of 60 pound, type III barrier.	500 to 999[4]
	A1-See Table XVI Mg-See Table XVII	A1-See Table XVI Mg-See Table XVII	Wirebound boxes (see fig. 10). Boxes shall be lined with two layers of 60 pound, type III barrier.	500 to 999[4]
	A1-See Table XVI Mg-See Table XVII	A1-See Table XVI Mg-See Table XVII	Double wall and triple wall corrugated fiberboard boxes (see fig. 11). Boxes shall be lined with two layers of 60 pound, type III barrier.[5]	500 to 999[4]

1 Where multiple layers of barriers are required, a single layer of barrier material which equals or exceeds the protective qualities of the multiple layers may be used in lieu thereof.

3 Sheet is 0.006 inch to 0.249 inch thick.
Plate is 0.250 inch and thicker.

4 When a specific number of pieces per box is desired the contract or order must so specify. The final box for any one item may contain one less than twice the specified number of pieces per box provided the weight of the final box does not exceed the specified standard net weight for that box.

5 Level A - Barrier liners may be omitted from double wall and triple wall fiberboard boxes when all exterior seams are sealed with tape, PPP-T-60 or equal.

Level B - Barrier liners or seam taping are not required, except when Style A End with Butt Joint Cover is used, taping of the joint as specified (see figure 11) will be accomplished.

6 Paper overlaid veneer, Wirebound, Solid or Corrugated Fiberboard Containers shall not be used for overseas shipments except when specified in contract or order.

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TABLE XIII. Preservation, Packaging and Packing for Level A (For level C, see 5.2.2)

Product	Preservation Level A (See 5.1)	Packaging Level A (See 5.3)	Packing Level A (See 5.3) (For Barriers see Table I)[1]	Standard Net Weight Per Container (See 5.5.12.1)
				Pounds
Sheet and Plate[3 & 4] (continued)				
Coiled Sheet	Oiled	None	Skids (see figs. 13-16). Skids shall be lined with one layer of 60 pound, type III and two layers of 30/80/30, type V barriers.	1,000 to 10,000
	Oiled	None	Wooden boxes (see figs. 1-4). Boxes shall be lined with two layers of 60 pound, type III barrier.	500 to 999
Circles	Al-See Table XVI Mg-See Table XVII	Al-See Table XVI Mg-See Table XVII	Skids (see figs. 13-16). Skids shall be lined with one layer of 60 pound, type III and two layers of 30/80/30, type V barriers.	Sheet: 1,000 to 4,000 Plate: 1,000 to 10,000
	Al-See Table XVI Mg-See Table XVII	Al-See Table XVI Mg-See table XVII	Wooden boxes (see figs. 1-4). Boxes shall be lined with two layers of 60 pound, type III barrier.	2,000 to 4,000
Roofing and siding sheet	None	None	Skids (see figs. 12-18). Skids shall be lined with two layers of 60 pound, type III barrier.	2,000 to 4,000
Floor and tread plate and abrasive tread plate	Al-See Table XVI Mg-See Table XVII	Al-See Table XVI Mg-See Table XVII	Bare metal strapped on skids (see figs. 15-18). Skids are not lined.	1,000 to 10,000
			Bare strapped bundles.	1,000

[1] Where multiple layers of barriers are required, a single layer of barrier material which equals or exceeds the protective qualities of the multiple layers may be used in lieu thereof.

[3] Sheet is 0.006 inch to 0.249 inch thick.
Plate is 0.250 inch and thicker.

[4] For overseas shipments of all sheet and plate aluminum or magnesium without a surface finish equal to or greater than a chromate or painted treatment shall have a preservative oil on both sides in accordance with paragraph 5.1.2. Interleaving paper shall not be used, and alclad material also requires oiling.

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TABLE XIII. Preservation, Packaging and Packing for Level A (For level C, see 5.2.2)

Product	Preservation Level A (See 5.1)	Packaging Level A (See 5.3)	P Packing Level A (See 5.3) (For Barriers see Table I) [1]	Standard Net Weight Per Container (See 5.5.12.1)
				Pounds
Structural Shapes (extruded and rolled)				
Less than 150 pounds per piece	A1-None Mg-Chrome pickled	None	Wooden boxes (see figs. 5-7).	300 to 1,000
150 pounds and over per piece	A1-None Mg-Chrome pickled	None	Double wall and triple wall corrugated fiberboard boxes (see 5.5.1.6 and table IIIa).	To 300
Tubular Products (drawn, extruded and welded)				
Coiled	A1-Oiled Mg-Oiled	Stagger wound coils shall be tied through the core in two places with twine or pressure sensitive tape. Pancake or level layer wound coils shall not be tied.	Crates (see fig. 22). Crates shall be lined with two layers of 60 pound, type III barrier.	100 to 700
Straight lengths	A1-Oiled Mg-Chrome pickled	When the wall thickness of tube is less than 2-1/2% of the outside diameter, boxes shall be lined with suitable corrugated fiberboard as a cushion.	Wooden boxes (see figs. 5-7). Boxes shall be lined with two layers of 60 pound, type III barrier.	300 to 1,000
			Double wall and triple wall corrugated fiberboard boxes (see para 5.5.1.6 and table IIIa).	To 300

[1] Where multiple layers of barriers are required, a single layer of barrier material which equals or exceeds the protective qualities of the multiple layers may be used in lieu thereof.

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TABLE XIII. Preservation, Packaging and Packing for Level A (For level C, see 5.2.2)

Product	Preservation Level A (See 5.1)	Packaging Level A (See 5.3)	Packing Level A (See 5.3) (For Barriers see Table I) [1]	Standard Net Weight Per Container (See 5.5.12.1) Pounds
Tubular Products (drawn, extruded and welded)				
ASA schedule pipe	In wooden boxes; A1-Oiled Mg-Chrome pickled In crates: A1-Oiled Mg-Chrome pickled	External threads shall be covered with suitable thread protectors. Pipe of different diameters and the same length shall be telescoped when approved by the procuring activity.	Wooden boxes (see figs. 5-7). Boxes shall be lined with two layers of 60 pound, type III barrier.	300 to 799
			Crates (see fig. 23). Crates shall be lined with one layer of 30/80/30, type V barrier.	800 to 4,000
Construction pipe	In wooden boxes: A1-Oiled In crates: A1-None Mg-Chrome pickled	Pipe of different diameters and the same length shall be telescoped when approved by the procuring activity.	Wooden boxes (see figs. 5-7). Boxes shall be lined with two layers of 60 pound, type III barrier.	300 to 799
			Crates (see fig. 23). Crates shall be lined with one layer of 30/80/30, type V barrier.	
Welding and Brazing Rod Coiled	None	None	Wrapped coils (see fig. 21). Wrapping shall be two layers of 30/30/30, , type IV barrier.	55 to 120
Straight lengths, 36"	None	Five pounds per fiber tube with metal ends. Ten tubes shall be packaged in double-faced, full-flap, slotted, fiberboard boxes per PPP-B-636, type CF, class domestic, except 275 pound test shall be used.	Wooden boxes (see figs. 1-4). Six fiberboard boxes shall be packed in each wooden box. Wooden boxes shall be lined with one layer of 60 pound, type III barrier.	300

[1] Where multiple layers of barriers are required, a single layer of barrier material which equals or exceeds the protective qualities of the multiple layers may be used in lieu thereof.

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TABLE XIII. Preservation, Packaging and Packing for Level A (For level C, see 5.2.2)

Product	Preservation Level A (See 5.1)	Packaging Level A (See 5.2)	Packing Level A (See 5.3) (For Barriers see table I) [1]	Standard Net Weight Per Container (See 5.5.12.1)
				Pounds
Welding and Brazing Rod (continued)				
Inert gas welding electrode	None	One pound per plastic spool, 10 pounds per fiber spool or 12-1/2 pounds per plastic spool. Spool dimensions shall be as specified in table XVIII. The periphery of electrode on spools shall be wrapped with minimum 0.00025-inch thick foil laminated to 30 pound kraft paper and held in place with tape. Each one-pound spool shall be packaged in a folding paperboard box per PPP-B-566. Each 10 or 12-1/2 pound spool shall be packaged in a corrugated fiberboard box per PPP-B-636, type CF (corrugated fiberboard), class weather resistant.	Fiberboard boxes per PPP-B-636, type CF (corrugated fiberboard), class weather resistant, grade V3c. All corners and seams of boxes, including manufacturer's joint, shall be sealed with tape per PPP-T-60, type I (water-vaporproof, waterproof), class I (exterior). Three 5/8-inch-by-0.015 inch flat steel straps, one lengthwise and two girthwise, shall be applied to each box. Twenty one-pound spools in paperboard boxes or five 10-pound spools, or four 12-1/2 pound spools in unit fiberboard boxes shall be packed in each fiberboard box.	One-pound spools - 20 10 pound spools - 50 12-1/2 pound spools - 50

[1] Where multiple layers of barriers are required, a single layer of barrier material which equals or exceeds the protective qualities of the multiple layers may be used in lieu thereof.

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TABLE XIV. Minimum Thickness of Flanges for Supporting Coiled Foil (See table XIII)

Weight Supported[1] Pounds	Flange Thickness Inch
Less than 100	5/8
100 and over	3/4 [2]

TABLE XV. Minimum Strap Sizes for Bundles and Skids of Ingot (See table XIII)

Type of package	Flat steel strapping Inch	Round Steel strapping Inch
Bundles	3/4 x 0.023	0.1055 (12 gage)
Skids	5/8 x 0.020	0.0800 (14 gage)

1 Several narrow coils may be suspended on one dowel.

2 Flanges 5/8 inch thick with 5/8 by 2-1/2 inch vertical cleats may be used in lieu of 3/4 inch thick flanges.

TABLE XVI - Preservation and Packaging of Aluminum Sheet and Plate[3] (See table XIII)

Product	Preservation and Packaging
Abrasive tread plate	None required.
Flat and tapered sheet and plate, and all circles	
Clad, all alloys	
Circles less than 12" diameter	None required.
Flat, tapered, and circles 12" diameter and over.	Interleaved.
Nonclad	
Circles less than 12" diameter	None required.
Flat, tapered, and circles 12" diameter and over.	
Heat-treatable alloys, all tempers[4]	Oiled.
Non heat-treatable alloys[4]	
Annealed	Interleaved.
Other than annealed	
0.064" thick and over	Interleaved.
Less than 0.064" thick	Oiled.
Floor and tread plate	Oiled.
Coiled sheet	Oiled.
Perforated Sheet	Oiled.
Roofing and siding sheet	None required.

3 Interleaving paper shall be Type I paper (see Table I), minimum 20 pound basis weight paper for all flat plate and sheet from and including .020 inches and thicker. Metal .019 and thinner may be interleaved with minimum 10 pound basis weight paper. The interleaving paper shall fully cover and extend beyond the ends of the sheet or plate a minimum of one (1) inch.

4 Heat-treatable alloys are the 2000, 6000 and 7000 groups.
Non heat-treatable alloys are the 1000, 3000 and 5000 groups.

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TABLE XVII. Preservation and Packaging of Magnesium Sheet and Plate[1] (See Table XIII)

Product	Preservation and Packaging[2]
Flat sheet and plate Less than 0.500 inch thick	Oiled or Oiled and interleaved or Chrome pickled and interleaved
0.500 inch and thicker	None or Oiled

1 Interleaving paper shall be as specified for aluminum sheet and plate. (See Note 3 table XVI).

2 Preservation and packaging desired must be specified in the contract or order.

TABLE XVIII. Dimensions for Spools of Inert Gas Welding Electrodes (see Table XIII)

	One Pound Plastic Spool		10 Pound Fiber Spool		12-1/2 Pound Plastic Spool	
Spool Part	Size Inches	Tolerance Inch	Size Inches	Tolerance Inch	Size Inches	Tolerance Inch
Width, Overall	1-3/4	≠ 1/32, -1/32	4	≠ 1/16, -1/16	4	≠ 1/16, -1/16
Flange diameter	4	≠ 0, -1/16	12 max.	---	12 max.	---
Arbor hole diameter	.630	≠ 0.005 - 0.000	2-1/16	≠ 1/16, -0	2-1/32	≠ 1/16, -0
Engaging hole diameter	none	---	7/16	≠0, -1/32	7/16	≠ 0, -1/32
Engaging hole location	none	---	1-3/4 off center	≠ 1/64, -1/64	1-3/4 off center	≠ 1/64, -1/64

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